

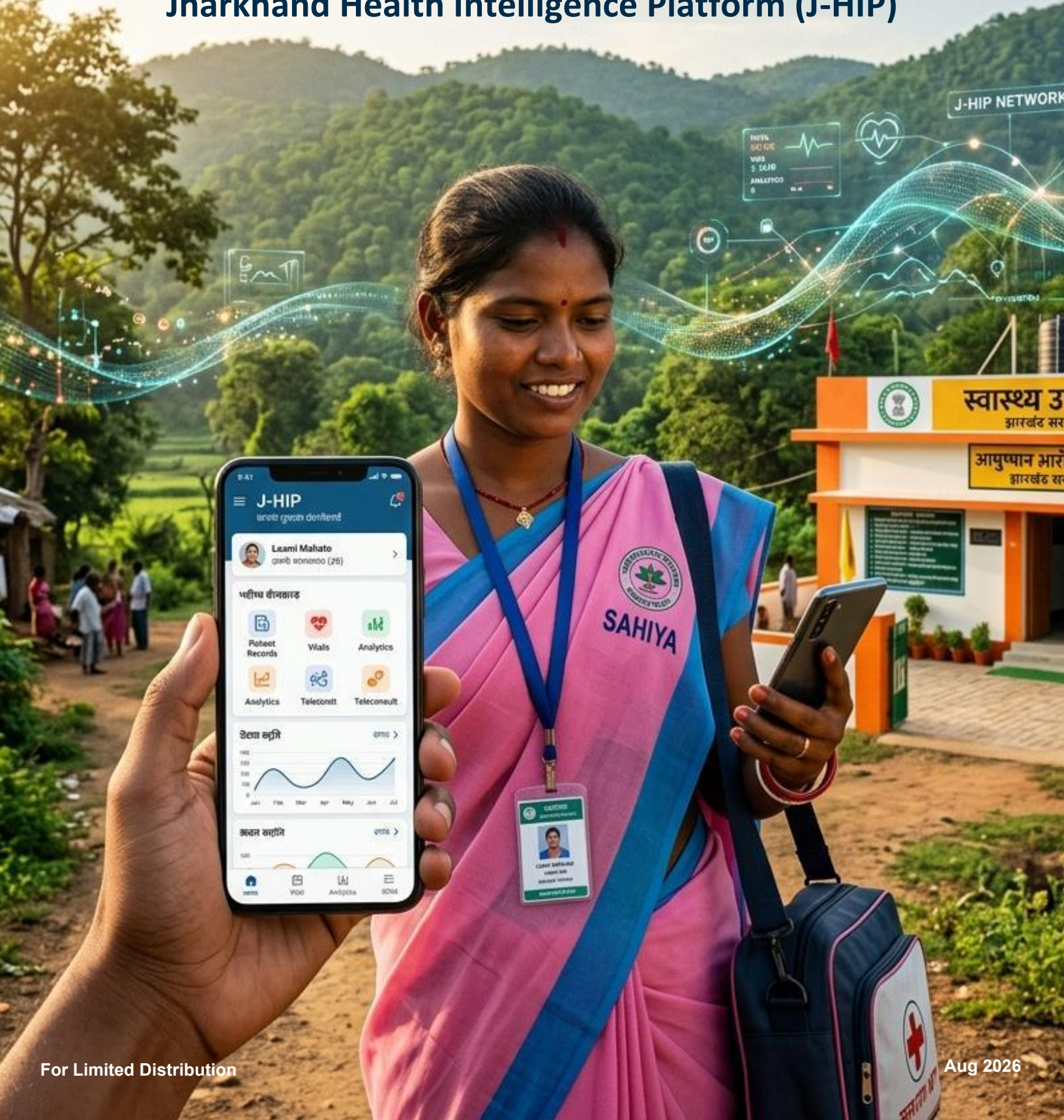


GOVERNMENT OF JHARKHAND

Department of Health, Medical Education &
Family Welfare

DETAILED PROJECT REPORT (DPR)

Jharkhand Health Intelligence Platform (J-HIP)



1. Table of Contents

1 Table of Contents.....	2
2. Project Overview.....	3
3. Executive Summary.....	4
4. Background	9
5. Project Description.....	15
6. Scope of the Project.....	18
7. Technical Requirements.....	28
8. Implementation Strategy	34
9. Financial Details	39
10. Risk Assessment & Mitigation.....	49
11. Sustainability.....	51
12. Annexures	54

2. Project Overview

Sponsoring Department	Department of Health, Medical Education & Family Welfare (DoHMEFW), Government of Jharkhand
Base Programme Cost	₹56.85 Cr
Total Programme Budget	₹77.15 Cr, inclusive of 10% contingency and 18% statutory GST, over 36 months
Compute Model	Hybrid architecture - State Data Centre (Ranchi) for primary hosting plus subsidised GPU cloud under the IndiaAI Mission for AI training/inference
Implementation Period	60 months: Months 1–3 design & build; Months 4–12 four-district pilot and validation; Months 13–60 statewide scale-up across 24 districts and O&M.

3. Executive Summary

3.1 Project Overview

The Jharkhand Health Intelligence Platform (J-HIP) is a state-wide, capital-light, AI-enabled health intelligence initiative proposed by the Department of Health, Medical Education & Family Welfare (DoHMEFW), Government of Jharkhand. Rather than building new hardware-heavy infrastructure, J-HIP layers artificial intelligence-based decision support, population health management, and multilingual care coordination on top of Jharkhand's existing, mature digital health ecosystem - including e-Shusrut, the Sahiya Portal/App, HMIS, DVDMS, PM-JAY TMS, Nikshay, and national platforms such as ABDM, eSanjeevani and IHIP.

The platform is designed to serve approximately 4.06 crore citizens across all 24 districts and ~66 lakh families, with particular emphasis on Jharkhand's 86 lakh tribal citizens, including eight Particularly Vulnerable Tribal Groups (PVTGs), and on mining-belt populations exposed to occupational respiratory hazards.

3.2 Context and Rationale

Jharkhand outperforms the national average on maternal mortality (MMR 82 vs 87, *Sample Registration System (SRS) 2022–24*) but lags on infant and under-five mortality (IMR 29 vs 25; U5MR 32 vs 29; SRS 2023). Structural and nutritional gaps persist: roughly 40% of children under five are stunted, 65.3% of women aged 15–49 are anaemic (*National Family Health Survey (NHS)-5, 2019–21*), and high rural CHC specialist vacancy. Geography and demography compound this — high forest cover (~29.8%, *India State of Forest Report (ISFR) 2023*) and a dispersed tribal population make service delivery harder, while the mining-belt economy adds occupational disease risk. Jharkhand is accordingly a designated high-prevalence priority state for both Sickle Cell Disease and silicosis - *Conference on Silicosis for Endemic States*.

Ground consultations across Bokaro, Ranchi and Khunti (14 sessions, 19 participants) found that Jharkhand's problem is not a shortage of digital applications, but fragmentation: frontline workers re-enter the same data across multiple portals, medical officers lack consolidated patient histories, and referral, discharge and follow-up information rarely reaches the responsible worker in time. J-HIP is therefore positioned as a common intelligence, interoperability and workflow layer across existing systems - not a parallel data-entry platform - converting already-collected data into beneficiary-level, closed-loop action.

This approach is reinforced by a strong digital foundation (1.8 crore digitised ABHA IDs, 40,000+ frontline workers already on the Sahiya App – *NITI for States*) and by demonstrated AI outcomes elsewhere in India - including Tamil Nadu's retinal-screening programme, Qure.ai's TB/silicosis chest X-ray triage, and ARMMAN's maternal-dropout prediction.

3.3 Vision and Objectives

J-HIP's vision is to transform state-wide public health delivery through an AI-driven, capital-light health intelligence platform that bridges physical infrastructure gaps and delivers equitable, continuum-based care, with special emphasis on tribal and other vulnerable populations.

- **Unify the digital health ecosystem:** establish a state-wide data lake and care-coordination engine connecting primary health centres to apex facilities across all 24 districts.
- **Enhance clinical and diagnostic decision support:** provide AI tools assisting medical officers with early screening for silicosis and TB, high-risk pregnancy progression, and neonatal distress.
- **Empower grassroots health workers:** equip ~40,000 Sahiyas with AI-prioritised task lists and vernacular audio checklists to improve home-visit compliance and follow-up.
- **Ensure scheme integrity and resource optimisation:** safeguard MASSY/PM-JAY funds through automated fraud detection and enable predictive maintenance of critical biomedical equipment.

3.4 Solution Pillars and Scope

J-HIP is organised around five strategic pillars comprising 16 core AI use cases:

- **Pillar 1: Population Health Management:** Eliminates fragmented manual registers by tracking community health programmes (post-discharge, chronic disease, maternal care, TB) and using AI to prioritize frontline workloads, ensuring timely care continuity and fewer readmissions.
- **Pillar 2: Tribal & Occupational Health:** Resolves unlinked health screenings in vulnerable populations by connecting occupational lung risks and sickle-cell anaemia directly to clinical care and integrating silicosis tracking with national TB registries.
- **Pillar 3: Clinical & Diagnostic Decision Support:** Tackles clinician burnout and delayed deterioration detection through point-of-care tools like high-risk pregnancy alerts, automated medical histories, and safety checks, boosting patient safety without replacing clinical judgment.
- **Pillar 4: Financial Integrity & Scheme Compliance:** Fixes revenue leaks and slow fund flows by automating multi-grant reconciliations, detecting insurance claims anomalies, monitoring regulatory compliance, and pre-auditing vouchers to accelerate grant disbursements and curb fraud.
- **Pillar 5: Hospital Performance Management:** Overcomes siloed operational data by tracking real-time patient volumes, equipment utility, workforce deployment, and

finances into an integrated facility scorecard, giving leadership instant visibility to optimize resources.

All AI outputs function strictly as Clinical Decision Support Advisories under a mandatory Human-in-the-Loop (HITL) protocol: no prescription, referral, admission, claim rejection or punitive action is executed autonomously - final authority remains with the licensed clinician or authorised officer. The scope explicitly excludes edge devices, diagnostic equipment procurement, and interventional clinical trials, and is designed to complement, not duplicate, the parallel NHM Dashboard RFP.

3.5 Technical Approach

The platform follows a four-stage pipeline - automated ingestion via APIs/SFTP/webhooks; a State Health Data Lake at the Jharkhand SDC with a 14-digit ABHA-based Master Patient Index; containerised AI inference, with heavy training workloads offloaded to subsidised IndiaAI Mission GPU compute; and role-based delivery through executive dashboards, Sahiya task lists, and a multilingual citizen IVR/WhatsApp/SMS gateway.

Data privacy is embedded by design in compliance with the DPDP Act 2023: ABDM-linked consent management, dynamic de-identification before model training, encryption at rest and in transit, role-based access control, immutable audit logging, and bi-annual CERT-In VAPT audits. Clinical safety is governed by explainable AI, mandatory bias and fairness testing across demographic sub-groups (with recalibration), and a multidisciplinary Clinical AI Safety Board reviewing model performance monthly.

3.6 Implementation Strategy

Deployment follows a risk-controlled, two-phase rollout over 60 months. Phase 1 (Months 3–12) pilots the platform in four representative districts - Ranchi (tertiary anchor/urban-rural mix), East Singhbhum (mining/silicosis belt), Khunti (Schedule V tribal district) and Dumka (Santhal Pargana tribal cluster) - followed by field validation and independent evaluation. Phase 2 (Month 12 onwards) scales the validated solution to all 24 districts, activating all 27 AI use cases statewide alongside 24/7 Operations & Maintenance for the remaining 48 months.

Capacity building follows a Cascade Training-of-Trainers model covering ~40,000 Sahiyas, ~8,000 ANMs/CHOs, ~2,800 medical officers and ~150 district leaders, supported by vernacular pictorial handbooks, video tutorials and accredited CME modules. Change management is anchored in community ownership through traditional tribal institutions and the “Abua” (“Our Own”) branding, with communication delivered in Santhali, Ho, Mundari, Kurukh, Khortha, Sadri and Hindi, including audio delivery for non-literate beneficiaries.

Payments to the System Integrator are milestone-linked during implementation (mobilisation, data-lake go-live, AI module integration, pilot acceptance, statewide go-live) and SLA-linked during the 48-month O&M phase, with a minimum core service uptime requirement of 99.5% and quarterly disaster-recovery drills.

3.7 Financial Summary

The total 5-year budget envelope is ₹77.15 crore (all-inclusive), built up from a base programme estimate of ₹56.85 crore plus a 10% contingency buffer, 5% executing-agency overhead, and 18% statutory GST. SI technical development and O&M manpower together account for the largest share of cost, with GPU compute (IndiaAI Mission subsidised infrastructure) and LLM token inference representing a comparatively small, usage-linked component. Core CPU infrastructure is hosted at zero incremental cost on the existing State Data Centre.

3.8 Risk Assessment

The DPR maintains a comprehensive risk register spanning clinical safety, data privacy, connectivity, clinical adoption, vendor lock-in, algorithmic/tribal bias, source-system integration, infrastructure load, and financial sustainability. The highest-severity risks are: false-negative AI outputs affecting clinical safety (mitigated through mandatory HITL, monthly Clinical AI Safety Board review, and override logging); DPDP Act non-compliance or data leakage (mitigated through end-to-end encryption, de-identification, and bi-annual CERT-In VAPT audits); and connectivity blind spots in tribal and forested blocks (mitigated through an offline-first mobile architecture with local SQLite caching and 2G/SMS fallback).

3.9 Sustainability and Expected Impact

J-HIP is designed for institutionalisation rather than pilot obsolescence: the State Health Data Lake, Master Patient Index and trained models are to be gazetted as sovereign assets of the Government of Jharkhand, with DSI heat-maps and follow-up metrics embedded into routine District Health Society reviews and Sahiya task lists linked to the existing ASHA incentive structure. The platform is built on an open-source-first stack (Python, PostgreSQL, Kubernetes, PyTorch) with full IP, source code and model weights transferred to the state on vendor exit, avoiding proprietary lock-in.

Indicative benefits include reduced preventable re-admissions under MASSY/PM-JAY, earlier interception of disease clusters through hyper-local surveillance, reduced catastrophic out-of-pocket expenditure for an estimated 3.5 lakh vulnerable households annually, and improved healthcare access for 86 lakh tribal citizens and PVTG communities through native-language voice AI and grassroots Sahiya enablement.

4. Background

Carved out as the 28th state of the Indian Union in November 2000, Jharkhand spans a geographic area of 79,716 square kilometres defined by rich mineral resources, dense forest cover and a vibrant demographic fabric. With a projected population of approximately 4.06 Crore across 24 districts, the state is home to indigenous citizens (26.21% Scheduled Tribes), including eight Particularly Vulnerable Tribal Groups (PVTGs).

Over the past decade, Jharkhand's Department of Health, Medical Education & Family Welfare has demonstrated progress in strengthening public health infrastructure, expanding social protection, and pioneering digital health solutions at the last mile.

4.1 Health Indicators

Jharkhand performs better than the national average on maternal mortality (MMR of 82 vs. 87 nationally, *Sample Registration System (SRS) 2022–24*) but continues to lag behind on infant mortality (IMR of 29 vs. 25 nationally, SRS 2023) and under-five mortality (32 vs. 29 nationally, SRS 2023). The state faces substantial structural and nutritional gaps. Current data indicates that approximately 40% of children under five are stunted and 65.3% of women aged 15–49 are anaemic (*National Family Health Survey-5 (NFHS-5), 2019–21*). Rural specialist vacancy in Community Health Centres (CHCs) remains at approximately 75%, with only a fraction of primary care facilities operating at optimal capacity.

Jharkhand has a forest cover of 29.81%, which is significantly higher than the national average of around 24–25% (*India State of Forest Report (ISFR) 2023*). The state is home to Particularly Vulnerable Tribal Groups (PVTGs) who largely reside in hard-to-reach geographies. This demographic profile introduces specific challenges. The Global Burden of Disease Study 2021 identifies India as having the second-highest global concentration of Sickle Cell Disease. Jharkhand ranks among the top states for its tribal population and is designated as a high-prevalence priority state under the National Sickle Cell Anaemia Elimination Mission. Additionally, due to regional mining activities, the National Human Rights Commission (NHRC) lists Jharkhand as one of the six high-risk and endemic states for silicosis, *NHRC Conference on Silicosis for Endemic States 2016*.

Nationally, Out-of-Pocket Expenditure (OOPE) accounts for 39.4% of Total Health Expenditure (*National Health Accounts (NHA), 2021–22*). NITI Aayog notes that the poor face severe hardship regarding treatment-related costs. Securing government-funded family insurance schemes against financial leakage is a strict priority. Recent national interventions highlight this risk, with the Centre rejecting 3.56 lakh fraudulent claims worth ₹643 crore and removing 1,114 hospitals from the Ayushman Bharat panel (*Rajya Sabha on Tuesday, 11 March 2025, by Sh Prataprao Jadhav*).

4.2 Jharkhand's Digital Health Applications

Jharkhand's public health architecture represents a mature digital ecosystem where all major national and state-level online portals are implemented. Below state-specific platforms directly address targeted operational workflows.

Major IT Application by Jharkhand - Department of Health, Medical Education & Family Welfare

Application / Platform	Sponsoring Cell / Agency	Operational Scope & Technical Modality	Key Data Assets Generated
e-Shusrut	HMIS Cell / C-DAC	Hospital Management Information System (HMIS) deployed at RIMS & selected DHs	Patient-level electronic medical records, clinical notes, discharge summaries
Sahiya Portal & App	ASHA Cell / NHM	Mobile application for ~40,000 Sahiyas to log village visits and claim incentive vouchers	Frontline village demographics, home visit logs, incentive claims
Malnutrition Treatment MIS	Child Health Cell / NRC	Web-based registry for Severe Acute Malnutrition (SAM) admissions across 70+ MTCs	Child anthropometric measurements, recovery curves, therapeutic feeding
e-Aushadhi (DVDMS)	DVDMS Cell / JSMSCL	Supply chain management for essential drugs and surgical consumables	District drug inventory, stockouts, batch expiry, consumption logs
Garima Jharkhand	PCPNDT Cell	Ultrasound clinic registration and Form-F compliance tracking for sex-ratio oversight	Diagnostic facility registrations, pregnant woman scan tracking
PM-JAY TMS	State Health Agency (SHA)	Transaction Management System for hospital claims pre-authorization and settlement	Inpatient admissions, treatment packages, diagnostic bills, claims data
Nikshay (NTEP)	State TB Cell / Central TB Division	Direct benefit transfer and clinical registry for Tuberculosis patients	Patient diagnostic status, sputum/CBNAAT results, adherence records

Below, core national platform with interportable architecture maintains a comprehensive digital health infrastructure.

Major IT Application by Central Government implemented in Jharkhand

Application / Platform	Sponsoring Cell / Agency	Operational Scope & Technical Modality	Key Data Assets Generated
eSanjeevani	MoHFW / C-DAC	National tele-consultation platform connecting AB-HWCs (spokes) to district/medical college specialist hubs.	Tele-consultation logs, digital e-prescriptions, clinical referral records.
RCH Portal (Reproductive & Child Health)	Maternal & Child Health Division, MoHFW	Web-based beneficiary tracking system for pregnant women and infants across all health facilities.	Pregnant women registrations, ANC/PNC tracking, child immunization schedules.
ANMOL (ANM Online)	MoHFW / State NHM	Tablet-based mobile application for Auxiliary Nurse Midwives (ANMs) to log real-time service delivery.	Door-to-door maternal-child health records, village-level service delivery logs.
ABDM / ABHA (Ayushman Bharat Digital Mission)	National Health Authority (NHA)	Digital health identity generation and state-wide registration of health facilities and medical practitioners.	Unique ABHA IDs, Health Facility Registry (HFR), Healthcare Professional Registry (HPR).
IHIP (Integrated Health Information Platform)	IDSP / NCDC / MoHFW	Real-time digital disease surveillance system for early epidemic response across all 24 districts.	Daily syndromic, presumptive, and laboratory-confirmed disease outbreak reports.
U-WIN Portal	Immunization Division, MoHFW	National digital platform tracking individual vaccinations under the Universal Immunization Programme (UIP).	Digital vaccination logs, automated SMS reminders, QR-coded e-vaccination certificates.
CPHC-NCD Portal	NCD Division, MoHFW	Longitudinal tracking platform for screening hypertension, diabetes, and oral/breast/cervical cancers at PHCs/HWCs.	Community-Based Assessment Checklist (CBAC) scores, screening logs, NCD treatment histories.
e-RaktKosh	Blood Cell, MoHFW / C-DAC	Centralized blood bank management system interconnecting licensed public and private blood banks.	Real-time blood component availability, donor registry, blood issue and collection logs.
eVIN (Electronic Vaccine Intelligence Network)	Immunization Division, MoHFW / UNDP	Digitized cold-chain stock management monitoring vaccine inventory and storage temperatures state-wide.	Real-time vaccine stock levels, cold-chain temperature logs, stockout alerts.
HMIS MoHFW	Statistics Division, MoHFW	Web-based health facility reporting portal aggregating monthly public health performance indicators.	Consolidated monthly facility performance data, key health service delivery statistics.

Source: Discussion with NHM and District health workers

4.3 Jharkhand's Readiness for AI-Enabled Healthcare

Jharkhand is uniquely positioned to implement an AI-enabled health tech solution due to three major structural shifts: a robust data foundation with 1.8 crore digitized Ayushman Bharat Health Account (ABHA) IDs, an established delivery rail of over 40,000 frontline workers already utilizing applications like the Sahiya app, and significant policy tailwinds from the ₹10,371.92 crore IndiaAI Mission. This structural readiness is reinforced by early AI implementation successes in other Indian states

The state's healthcare delivery continuum -ranging from prevention and diagnostics to longitudinal management and population surveillance- is supported by mature, ready-to-scale digital infrastructure. Rather than requiring entirely new data systems, Jharkhand's strategy focuses on integrating AI decision-support layers directly into existing platforms, such as the state-owned Sahiya App, national systems like the Integrated Health Information Platform (IHIP), and eSanjeevani telemedicine networks. By leveraging this established technological backbone- including linking PM-JAY and MASSY claims with longitudinal ABHA records and DVDMS drug inventory data- Jharkhand can seamlessly deploy predictive analytics and protocol-driven care to its rural and tribal populations without the burden of additional device costs.

4.4 Existing National AI Solutions in India

Nationally, Artificial Intelligence is transitioning from experimental proofs-of-concept to mission-critical public health infrastructure. Below is a structured analysis of relevant public and private implementations across India:

Solution / Initiative	Implementing Entity / State	AI/ML Modality & Core Architecture	Remarks
Tamil Nadu TNeGA AI	Tamil Nadu e-Governance Agency	AI retinal screening for diabetic retinopathy at primary health centres	Validates point-of-care AI decision support for rural non-specialist health cadres
Qure.ai Chest X-Ray AI	NHM Assam, Odisha, Maharashtra (TB Mission)	Deep learning computer vision (qXR) detecting TB, silicosis & pulmonary nodules	Models leveraged for mining belt Silicosis and Coal Workers' Pneumoconiosis screening
ARMMAN mMitra AI	ARMMAN & Google Research (Maharashtra/PAN-India)	Predictive engagement algorithms assessing dropout risk from maternal voice calls	Direct foundation for J-HIP Population Health Management maternal adherence voice outreach
Wadhvani AI (ORCHID)	NHM (Pest & Maternal Health)	Computer vision for infant anthropometry & risk stratification for low-birthweight	Applied in J-HIP Neonatal Sepsis & Distress Early Warning module

Solution / Initiative	Implementing Entity / State	AI/ML Modality & Core Architecture	Remarks
Piramal Swasthya AI	Govt of Assam / Piramal Swasthya	High-Risk Pregnancy (HRP) progression scoring integrated into 104 health helplines	Direct inspiration for J-HIP maternal progression scoring and 104 tele-triage integration
Swasth AI Chatbot	Swasth Alliance / Private Providers	Conversational NLP for chronic NCD medication adherence & lifestyle coaching	Adaptable for J-HIP multi-dialect conversational engine in Santhali and Ho
NITI Aayog / Apollo AI	NITI Aayog Pilot / Apollo Hospitals	Cardiovascular Risk Score algorithm tailored to Indian genetic profiles	Integrated into J-HIP Health Compliance & Risk Management (HCRM) module

4.5 Stakeholder Consultations

Ground research, inform of Stake Holder Interviews (Annexure A) the proposed Jharkhand Health Intelligence Platform (J-HIP) was conducted across selected facilities and blocks in Bokaro, Ranchi and Khunti through qualitative consultations with medical officers, frontline health workers including ANMs, Sahiyas and CHOs, Block Programme Managers, Block Data Managers, and vendors supporting state health applications. The exercise covered 14 consultation records involving 19 participants and was supplemented by a vendor assessment of selected health applications. Stakeholders highlighted a fragmented digital ecosystem characterised by multiple programme-specific applications, repeated data entry, dependence on paper registers and spreadsheets, limited longitudinal patient visibility, weak referral and discharge follow-up, beneficiary-matching problems, substantial reporting workload, language barriers and limited integration between clinical, operational and medicine-availability information. Frontline workers particularly emphasised the absence of actionable beneficiary worklists and timely follow-up information.

The findings support J-HIP as a common intelligence, interoperability and workflow layer across existing health applications, rather than another standalone system requiring parallel data entry. Foundational capabilities should include interoperable APIs, common identifiers, event-based notifications, offline synchronisation, role-based access and reliable data-exchange mechanisms. These should be complemented by AI-enabled capabilities such as beneficiary prioritisation, risk stratification, longitudinal record summarisation, entity-resolution assistance, anomaly detection, multilingual voice interaction, automated reporting, demand forecasting and guideline-linked clinical decision support. Vendor responses indicate varying levels of readiness across applications, requiring independent technical validation before implementation. As the consultations were purposive and limited to three districts, the findings should be treated as operational evidence rather than statistically representative statewide conclusions.

4.6 Gap Analysis and Need Assessment

Jharkhand has established a substantial digital-health ecosystem comprising national and state applications for hospital management, maternal and child health, immunisation, tuberculosis, non-communicable diseases, medicine management, frontline-worker reporting and health-protection schemes. Ground consultations, however, show that these systems largely function independently. Health workers repeatedly enter the same information across applications and registers, while medical officers lack consolidated patient histories. Discharge information, referral outcomes and programme-specific follow-up requirements do not consistently reach the responsible frontline worker. Consequently, beneficiary prioritisation, due-list preparation, data reconciliation and follow-up continue to depend on diaries, spreadsheets and telephone calls.

As per the study and ground research, the principal gap is therefore not the absence of applications or dashboards, but the limited conversion of available data into beneficiary-level actions and closed-loop care. Important unmet requirements include post-discharge monitoring, chronic-disease care-gap identification, maternal and newborn continuity, immunisation and TB adherence, occupational respiratory-health follow-up, sickle-cell and anaemia care completion, clinical safety support, claims scrutiny and scheme-entitlement identification. These gaps are particularly significant for tribal and PVTG communities, mining-belt workers, remote households and patients requiring care across multiple facilities.

The proposed use cases address these unmet workflows without duplicating the NHM Dashboard RFP. Common integration, reporting, outbreak surveillance, multilingual interfaces, bed management, workforce planning and inventory forecasting will remain within the RFP. J-HIP will instead use those shared capabilities to deliver distinct, explainable and human-supervised intelligence for continuity of care, clinical support, financial protection and regulatory oversight.

5. Project Description

The Jharkhand Health Intelligence Platform (J-HIP) is a state-wide public health initiative spearheaded by the Department of Health, Medical Education & Family Welfare (DoHMEFW), Government of Jharkhand. Designed to serve a target population of ~4.06 Crore citizens across all 24 districts of Jharkhand, J-HIP overcomes physical infrastructure constraints by utilizing Artificial Intelligence (AI) for decision support, population health management, and automated multilingual care coordination.

5.1 Vision

To transform state-wide public health delivery across Jharkhand by establishing an AI-driven, capital-light health intelligence platform. The initiative aims to bridge physical healthcare infrastructure gaps and deliver equitable, continuum-based care to all citizens, with specialized emphasis on vulnerable tribal communities and high-risk populations.

5.2 Objectives

- **Unify the Digital Health Ecosystem:** Establish a state-wide data lake and care-coordination engine connecting primary health centers to apex facilities across all 24 districts.
- **Enhance Clinical & Diagnostic Decision Support:** Provide AI-driven decision tools to assist medical officers with early screening for occupational health hazards (such as silicosis and tuberculosis), high-risk pregnancy progression, and neonatal distress.
- **Empower Grassroots Health Workers:** Equip frontline Sahiyas with localized task orchestration tools and vernacular audio checklists in native dialects to improve home-visit compliance and post-discharge follow-ups.
- **Ensure Scheme Integrity & Resource Optimization:** Safeguard public health funds through automated fraud detection in health insurance schemes while optimizing budget flows and predictive maintenance for critical biomedical equipment.

5.3 Solution Pillars

The Jharkhand Health Intelligence Platform (J-HIP) is organized into 5 Core Strategic Pillars:

Pillar 1: Population Health Management

Pillar 1 solves the problem of fragmented follow-ups and reliance on manual registers by tracking patients across community health programmes—such as post-discharge care, chronic diseases, maternal-child health, and TB—while using AI to prioritize frontline worker workloads. Its core function is to automate risk stratification and scheduling, and the envisaged benefit is ensuring

timely care continuity, reducing preventable readmissions, and giving supervisors clear visibility over high-risk or overdue patients.

Pillar 2: Tribal & Occupational Health

Pillar 2 addresses the critical gap where mobile or vulnerable populations undergo initial health screenings that never translate into actual clinical treatment or follow-up. Its core function is to connect occupational respiratory risks and sickle-cell anaemia screenings directly to confirmatory care pathways while linking dust-related lung diseases like silicosis with national TB (Nikshay) tracking. The envisaged benefit is a unified management approach for marginalized tribal and mining communities that ensures no diagnosed patient falls through the cracks of the healthcare system.

Pillar 3: Clinical & Diagnostic Decision Support

Pillar 3 tackles clinician burnout, manual history-reconstruction errors, and delayed responses to clinical deterioration in high-volume settings. Its core function is to deliver point-of-care support tools—including high-risk pregnancy alerts, neonatal sepsis early warnings, AI-generated patient summaries, medication safety checks, and closed-loop referral tracking. The envisaged benefit is enhanced patient safety, strict protocol adherence, and reduced administrative burdens for medical officers without replacing human clinical judgment.

Pillar 4: Financial Integrity & Scheme Compliance

Pillar 4 solves the systemic issues of revenue leaks, delayed fund velocities, insurance fraud, and error-prone manual pre-audits. Its core function is to automate multi-grant financial reconciliations, detect anomalies in MASSY/PM-JAY claims, track PC&PNDT compliance, and pre-audit vendor vouchers. The envisaged benefit is increased fraud-recovery yields, accelerated central and state grant disbursements by weeks, prevented audit objections, and the seamless redirection of eligible citizens into cashless healthcare pathways.

Pillar 5: Hospital Performance Management

Pillar 5 addresses the lack of real-time operational visibility caused by siloed departmental data and static, unreliable asset registers. Its core function is to track actual patient volumes, equipment functionality, workforce deployments, and financial flows, combining them into an integrated monthly facility scorecard. The envisaged benefit is that state and district leadership gain a single, verified executive dashboard to quickly identify underperforming facilities, optimize resource allocation, and drive administrative accountability.

5.4 Key Stakeholders

Stakeholder Group	Role & Interest	Engagement Modality
-------------------	-----------------	---------------------

DoHMEFW & State Health Department	Policy owner; budget holder; programme steward	Steering committees; policy notes; regular reviews
District Health Societies & Civil Surgeons	Local implementation; facility oversight	District cells; training; dashboards
ASHAs, ANMs, CHOs (Frontline Workers)	Primary end-users; data collectors	ToT training; user feedback loops; micro-learning
Doctors & Specialists	Clinical validators; AI-output reviewers	Clinical Validation Committee; CMEs; co-design
Patients & Communities	Beneficiaries; data principals	IEC campaigns; community meetings; grievance channels
Finance & IT Departments	Procurement; infrastructure; fiscal control	Joint working groups; appraisal processes
JMHIDPCL	Procurement & contract management	RFP design; bid evaluation; SLA monitoring
Vendors & System Integrators	Solution development; operations	Contractual reviews; sandbox evaluation; SLA reporting
C-DAC / NIC	Govt. IT systems; HMIS integration	Technical committees; API integration planning
NGOs, CBOs & Academia	Implementation support; community reach; evaluation	Partnership MoUs; independent evaluations

6. Scope of the Project

6.1 Functional Requirements

Pillar 1: Population Health Management

Use case	What and whom are being targeted	Envisaged AI use case	Envisaged benefit
1.1 Post-discharge care and complication-risk management	Patients discharged from public and empanelled hospitals, particularly MASSY and PM-JAY beneficiaries; Sahiyas, ANMs, CHOs, care coordinators and medical officers responsible for follow-up	Analyse the available discharge record, diagnosis, procedure, comorbidities and prior utilisation; classify follow-up risk; prepare a condition-appropriate follow-up schedule; analyse patient responses for red-flag symptoms and escalate exceptions to the responsible worker or clinician	Timely digital handover from hospital to community care; consistent follow-up instead of separate registers and telephone calls; visible missed contacts; earlier clinical review of reported complications; auditable follow-up completion and readmission data
1.2 Chronic-disease care-gap and multi-disease management	Patients with diabetes, hypertension, cardiovascular disease, chronic kidney disease and other chronic or multiple conditions; NCD teams, CHOs and medical officers	Use longitudinal visits, investigations, vitals, medicines and previous admissions to identify overdue tests, missed reviews, refill gaps and rising clinical risk; generate a prioritised care plan for professional review	High-risk and overdue patients become visible before the next routine OPD visit; NCD teams can focus follow-up on identified gaps; medicine, test and review compliance can be tracked through a single care pathway
1.3 Maternal, postnatal and newborn continuity tracking	Pregnant women, recently delivered mothers and newborns who miss or delay scheduled services; ANMs, Sahiyas and maternal-health teams	Analyse the sequence of registrations, ANC visits, investigations, delivery, discharge and postnatal events to identify likely care discontinuity; prioritise overdue mother–newborn pairs for follow-up	Automatically prepared due and overdue lists; faster notification of delivery and discharge; fewer cases dependent on personal diaries; traceable completion of ANC, postnatal and newborn follow-up
1.4 SAM/MTC discharge and nutritional-relapse risk management	Children admitted to or discharged from Malnutrition Treatment Centres and children with severe acute malnutrition; MTC teams, ANMs, Sahiyas and child-health teams	Analyse weight-gain trajectory, discharge status, comorbidities, missed follow-up and subsequent growth measurements to identify children needing early review	MTC discharge does not end the care pathway; missed follow-up and faltering recovery become visible; community workers receive child-specific follow-up tasks; programme teams can track recovery completion
1.5 Immunisation defaulter and dropout recovery	Children and pregnant women with due, overdue or partially completed vaccination schedules;	Use U-WIN and related service history to identify missed doses and estimate the likelihood of future dropout;	Due lists no longer have to be reconstructed manually after each session; workers can focus first on overdue and

	ANMs, Sahiyas and immunisation programme managers	generate beneficiary-level follow-up priorities	high-dropout-risk beneficiaries; recovery of missed doses becomes measurable
1.6 TB treatment-adherence and post-treatment follow-up	People registered in Nikshay who miss treatment events, investigations or reviews, including patients requiring post-treatment follow-up; TB programme staff, treatment supporters and frontline workers	Analyse treatment events, missed medicine collection, investigation results, previous interruptions and contact outcomes to flag patients at risk of treatment interruption or loss to follow-up	Treatment supporters receive prioritised follow-up lists; missed treatment events are identified earlier; unresolved cases and completed follow-up are visible to programme managers
1.7 AI-guided frontline visit prioritisation	Sahiyas, ANMs and CHOs managing more beneficiaries than can be visited in a day; beneficiaries with discharge, maternal, immunisation, nutrition, TB and NCD follow-up requirements	Combine approved risk flags and programme deadlines into a ranked daily or weekly worklist showing whom to visit, the reason for the visit, the required action and the completion deadline	Replaces manual selection from multiple registers; urgent and overdue cases are acted upon first; supervisors can see pending, completed and escalated work; field workload becomes more structured and accountable

* Implementation of Above use-cases (1.1-1.7) will be supported by **Patient Risk scoring** to prioritize the high risk patients

Pillar 2: Tribal & Occupational Health

Use case	What and whom are being targeted	Envisaged AI use case	Envisaged benefit
2.1 Occupational respiratory-risk and follow-up	Silica, stone-crusher, mining and coal workers exposed to occupational dust; occupational-health teams, mobile medical units and clinicians in mining districts	Use occupational exposure, symptoms, spirometry, previous diagnoses, treatment and admission history to rank workers for clinical screening, confirmation and long-term review. AI interpretation of chest X-rays is excluded because medical-report and radiology analysis falls under the RFP	Limited screening and clinical capacity can be directed first to workers with stronger risk indicators; occupational exposure and follow-up remain connected across illness episodes; missed reviews and treatment gaps become visible
2.2 Silicosis/CWP-TB coordinated care and Nikshay linkage	Workers with confirmed or suspected silicosis, coal workers' pneumoconiosis or other dust-related disease who also require TB testing or treatment	Match authorised occupational-health records with Nikshay events; identify workers without required TB testing, unresolved results, treatment interruption or follow-up after treatment;	Occupational lung disease and TB are managed as a connected pathway; eligible workers are referred for testing without creating a parallel TB registry; programme teams can see whether testing, treatment

		create programme-specific actions	and follow-up were completed
2.3 Sickle-cell and anaemia screening-to-care completion	Scheduled Tribe and PVTG populations screened for sickle-cell disease, sickle trait or anaemia; screening teams, counsellors, ANMs, CHOs and designated treatment facilities	Analyse screening, confirmatory test, counselling, treatment and follow-up events to identify incomplete pathways and prioritise people requiring confirmation or clinical review	Screening results are converted into completed care pathways; confirmatory tests and clinical follow-up are less likely to remain pending; district teams receive a clear view of unresolved cases. Counselling remains voluntary, confidential and clinician-led

* Regional and tribal-language voice support is a mandatory shared accessibility service under the NHM Dashboard RFP. It should support all three use cases but should not be procured again as a separate J-HIP use case.

Pillar 3: Clinical & Diagnostic Decision Support

Use case	What and whom are being targeted	Envisaged AI use case	Envisaged benefit
3.1 High-risk pregnancy and escalation support	Pregnant women with clinical or obstetric risk indicators; ANMs, obstetric teams, medical officers and referral units	Analyse maternal parameters, obstetric history, investigation results and their change over time to identify increasing risk of complications; present explainable risk factors and required protocol review to the authorised clinician	Earlier review of women whose risk is increasing; consistent application of approved maternal-care protocols; traceable escalation. AI does not order mandatory investigations, determine delivery mode or dispatch an ambulance autonomously
3.2 Neonatal and sepsis early-warning support	Newborns managed in SNCUs and NBSUs; neonatal nurses and clinicians	Continuously evaluate available vital signs, clinical observations, birth characteristics and laboratory trends to identify deterioration patterns and generate a review alert	Deterioration indicators are assessed consistently between manual rounds; clinicians receive a consolidated trend and reason for the alert; acknowledgement and response time can be monitored
3.3 AI-assisted patient summary	Medical officers who currently reconstruct history from paper records, family recollection and separate applications; patients with records across e-Sushrut, ABHA and programme systems	Summarise available diagnoses, medicines, allergies, investigations, admissions, referrals and unresolved follow-up items while retaining links to the source record	Less consultation time spent reconstructing history; significant previous conditions and medicines are less likely to be overlooked; clinicians can verify every summarised item against its source. Existing CARE-HMIS summary functionality should be reused rather than duplicated

Use case	What and whom are being targeted	Envisaged AI use case	Envisaged benefit
3.4 Medication-safety and guideline support	Medical officers and patients receiving treatment in high-volume PHC, CHC and hospital settings	Compare proposed medicines with recorded allergies, diagnoses, laboratory results, pregnancy status, current medicines and approved treatment guidelines; flag potential interactions, duplications, contraindications or protocol deviations for clinician review	Provides a consistent prescription-safety check in settings with limited specialist support; makes the reason for each warning visible; supports safer review without allowing the system to prescribe or change treatment
3.5 Inpatient deterioration and discharge-readiness support	Hospitalised patients with increasing clinical risk or a likelihood of an unsafe or incomplete discharge; ward clinicians and discharge teams	Analyse trends in observations, investigations, length of stay, comorbidities and previous utilisation; flag patients needing senior review and identify unresolved discharge-plan requirements	Enables structured review of deteriorating patients; incomplete discharge instructions, medicines or follow-up plans can be addressed before discharge; creates a direct link with post-discharge use case 1.1
3.6 Closed-loop referral outcome tracking	Patients referred between AAMs, PHCs, CHCs, district hospitals and tertiary facilities; referring and receiving clinicians, Sahiyas and ANMs	Analyse expected referral events and identify missing arrival, consultation, admission, treatment or discharge confirmation; prioritise referrals likely to remain incomplete and summarise the outcome for the referring team	Replaces informal feedback through telephone calls or ASHA communication; the referring clinician can see whether care was received; incomplete referrals are followed up. Selection of hospitals, beds or ambulance routing remains within the RFP

Pillar 4: Financial Integrity & Scheme Compliance

Use case	What and whom are being targeted	Envisaged AI use case	Envisaged benefit
4.1 Executive Health Financial Command & Multi-Grant Reconciliation Dashboard	Senior state and district health officials, State Finance Directorate, Treasury & State Nodal Agencies	Integrates, standardizes, and cross-reconciles multi-source financial feeds across State Treasury (Kuber IFMS), Central Schemes (PFMS/SNA), and off-budget funds (DMFT, XV Finance Commission). Stitches transactions via a Unified Project/Facility Identifier to provide a real-time, consolidated master ledger and fund-velocity bottleneck pipeline.	Benefit: Gives state leadership a unified view of total available funds, inter-system bottlenecks, and vendor-level commitments across distinct revenue streams. Safeguard: Operates as a non-invasive, read-only analytics middleware; does not alter master accounting entries in source treasury portals.
4.2 MASSY & PM-JAY Claims Anomaly Detection	Suspicious pre-authorization/claim patterns across packages & providers;	Analyzes claims for duplicate billing, unbundled packages,	Benefit: Directs limited audit capacity toward high-risk claims, creating a clear audit

	Medical Auditors & Scheme Administrators (JSAS)	upcoding, atypical length of stay, repeated admissions, and provider-level billing outliers.	trail and increasing fraud-recovery yields. Safeguard: Generates explainable audit queues; system does not automatically reject claims or initiate punitive action.
4.3 Scheme Entitlement & Coverage Gap Identification	Patients/households eligible for MASSY or PM-JAY paying out-of-pocket; Scheme Desks & Beneficiary Support Teams	Cross-references real-time hospitalization inputs against ration card and SECC data to surface unlinked, eligible beneficiaries and incomplete applications.	Benefit: Directs eligible patients to cashless pathways prior to avoidable out-of-pocket spending, driving full scheme coverage. Safeguard: Entitlement decisions remain strictly with authorized government officials.
4.4 PC&PNDT Compliance Risk & Inspection Prioritization	Diagnostic facilities, genetic clinics, and ultrasound centers registered in Garima Jharkhand; State/District Appropriate Authorities	Evaluates duplicate records, Form-F reporting lags, registration gaps, and historical compliance trends to compute a dynamic facility risk score.	Benefit: Optimizes limited inspection bandwidth by focusing physical audits on high-risk diagnostic centers. Safeguard: System acts as a risk-ranking tool; does not determine statutory violations or execute legal enforcement automatically.
4.5 Pre-Audit & AC/DC Bill Compliance Agent	Emergency disbursements, contingent bills, and vendor vouchers; DDOs (Civil Surgeons, Medical Superintendents) & Treasury Officers	Scans invoices, GSTIN entries, and subvouchers prior to treasury submission, checking against Jharkhand Financial Rules (JFR) threshold limits and split-billing rules.	Benefit: Catches format errors and missing mandatory fields early, preventing Accountant General (AG) audit objections and un-cleared Abstract Contingent (AC) bills. Safeguard: Pre-audit assistant highlights risks; payment release authority remains exclusively with the DDO.
4.6 Multi-Grant Fund Velocity & UC Triggers	Capital infrastructure funds across Treasury (Kuber), SNA (PFMS), and off-budget accounts (DMFT/XV FC); District Program Managers & Finance Directorate	Tracks real-time expenditure velocity across combined grant streams and auto-populates Utilization Certificate (UC) drafts when disbursement thresholds (e.g., 70–75%) are reached.	Benefit: Prevents funds from sitting idle in district bank accounts and unlocks central/state grant tranches 30–45 days faster. Safeguard: Generates pre-filled draft forms (e.g., JTC Form 62/63 equivalent); requires physical verification and signature by the DDO before filing.

Pillar 5: Hospital Performance Management

Use case	What and whom are being targeted	Envisaged AI use case	Envisaged benefit
5.1 Facility-wise patient volume and speciality-service tracking	OPD and IPD patients across PHCs, CHCs, sub-divisional and district hospitals; State Leadership, Civil Surgeons and District Health Societies	Consume HMIS/e-Shusrut footfall and encounter data; classify visits by speciality/service line (e.g. ENT, Ophthalmology, Orthopaedics, Gynaecology) using facility service-mapping and diagnosis codes; distinguish new, follow-up and referred patients; flag facilities reporting zero or implausible OPD activity	State Leadership gets a single verified view of how many patients are being treated, where, and for what services, without a parallel data-entry system; facilities silently under-reporting or not functioning become visible for administrative action
5.2 Diagnostic and biomedical equipment utilisation and non-reporting-asset flagging	Registered biomedical and diagnostic equipment (X-ray, USG, CT, dialysis, ventilators, laboratory analysers) at all facilities; biomedical engineers, Civil Surgeons and facility administrators	Cross-check the equipment asset register against actual usage/report logs generated by the equipment or the concerned department; flag equipment recorded as available but generating no reports over a defined period (possible non-functional, missing or unused asset), and equipment absent from the register despite departmental activity suggesting its use	Converts the equipment register from a static inventory into a living, verified list; equipment shown on paper but not functioning, or not physically present, is surfaced for inspection rather than being discovered during an audit or a crisis
5.3 Hospital workforce (HR) availability and deployment tracking	Sanctioned versus in-position medical, nursing, paramedical and diagnostic staff at each facility; District Health Societies, Civil Surgeons and State HR Cell	Reconcile HRIS/attendance data with sanctioned posts and roster requirements by facility and department; identify chronic vacancy, mismatched deployment (e.g. specialists posted where OPD volume does not require them) and facilities running services without the requisite staff on record	Specialist and staff vacancy, already flagged qualitatively in this DPR (~75% rural CHC specialist vacancy), becomes a facility-level, continuously updated figure that can be acted on rather than a periodic survey finding
5.4 Facility and scheme financial-flow tracking	Budget allocation, expenditure and fund utilisation at facility, district and scheme level (State Budget, NHM PIP, MASSY/PM-JAY); Finance Department, DoHMEFW and District Health Societies	Track sanctioned budget against actual utilisation and pending liabilities by facility, scheme and head of account; flag significant unspent balances, delayed reimbursements, or spending that is not reconcilable with reported patient volumes or claims	Financial position of the health system becomes visible alongside clinical and operational data rather than in a separate finance-only reporting cycle, supporting more timely budget reappropriation and PIP planning
5.5 Integrated facility performance scorecard	All public health facilities from PHC to tertiary level; Chief Secretary, Principal Secretary, Civil Surgeons and District Health Societies	Combine the outputs of 5.1–5.4 with existing HMIS indicators into a single weighted facility scorecard covering patient services, workforce, diagnostics/equipment and financials; generate a plain-	Replaces separate HR, financial, equipment and patient-care reviews with one composite, comparable facility score; underperforming facilities are identified early for administrative

Use case	What and whom are being targeted	Envisaged AI use case	Envisaged benefit
		language narrative summary of each facility's standing and month-on-month movement for executive review	attention, and improvement can be tracked over time

6.2 Scope of Work – System Integrator

The selected agency/System Integrator (SI) will provide end-to-end advisory, technical, and implementation support to the Department of Health, Medical Education & Family Welfare (DoHMEFW), Government of Jharkhand, for the development, rollout, and adoption of the Jharkhand Health Intelligence Platform (J-HIP). The scope involves executing a capital-light, AI-driven digital health ecosystem over a 60-month engagement (12 months of development/rollout and 48 months of Operations & Maintenance).

The scope includes, but is not limited to, the following core functional areas:

1. Assessment, Planning & Architecture Design

- Conduct process mapping, define the Minimum Viable Product (MVP), and execute a 60-month phased rollout roadmap spanning 24 districts.
- Design a hybrid technical architecture utilizing the Jharkhand State Data Centre (SDC) for primary data hosting and secure APIs to leveraging subsidized IndiaAI Mission GPU cloud compute. The architecture must support ultra-low-latency REST and WebSocket APIs to handle concurrent, population-scale generative AI inference and real-time voice translation requests without bottlenecking the core application.
- Develop a unified State Health Data Lake with optimized columnar tables (Apache Parquet / Delta Lake) and a Master Patient Index (MPI) leveraging 14-digit ABHA IDs for deduplication and longitudinal record linkage.
- The Vendor will support Buyer in establishing and maintaining the AI Cloud as required and maintain the billing. Any cost incurred in this account shall be reimbursed to the vendor at actuals. The vendor shall ensure that the total usage cost is within the budget approved by the buyer.

2. Application & Dashboard Development

- Design and develop Core platform Architecture for ingestion of data and provide backend services indicatively consisting of below modules:
Core 1: State Health Data Lake and data engineering pipelines.
Core 2: Master Patient Index (MPI) and ABHA linkage engine.
Core 3: Clinically compliant FHIR R4 interoperability hub and API security gateway.

Core 4: Unified health mobile application, analytics command centre, and web portal.

- Design and develop a mobile application for frontline workers (Sahiyas/ANMs) featuring offline-first architecture with localized SQLite caching to ensure operational continuity in zero-connectivity tribal regions.
- Implement an intelligent engine that pushes AI-prioritized daily home-visit task lists directly to the Sahiya App.
- Embed audio-guided visit checklists and conversational interactions in native tribal languages (Santhali, Ho, Mundari, Kurukh, Khortha, Sadri) natively into the app, leveraging the leveraging national sovereign AI translation and speech Digital Public Infrastructure (DPI) .
- Build role-based web dashboards for State Leadership, Civil Surgeons, and District Health Societies.
- Implement advanced Optical Character Recognition (OCR) and Text-to-Speech (TTS) capabilities within the mobile application, allowing frontline workers to capture images of printed medical documents, legacy registers, or ID cards and have the extracted text read aloud in their native dialect to support low-literacy users.
- Integrate real-time epidemiological Disease Severity Index (DSI) heat-maps to track hospital utilization, preventable readmission clusters, and chronic disease prevalence at the block and district levels.
- Integrate high-accuracy Automatic Speech Recognition (ASR) optimized for noisy field environments and 'code-switching' (e.g., seamlessly understanding sentences that mix Hindi, English, and regional tribal dialects) to enable hands-free, voice-driven data entry for Sahiyas.

3. Core AI / ML Module Development

Design, fine-tune, and deploy the 27 core Artificial Intelligence use cases (mentioned in functional requirement) across the following strategic pillars:

- **Population Health Management (Pillar 1):** Develop automated Post-Discharge Care Management (PDCM) follow-ups via IVR/WhatsApp powered by natural, emotion-aware Text-to-Speech (TTS) synthesis and multi-turn conversational AI (enabling human like dialogue) and Chronic Disease Health Compliance & Risk Management (HCRM) stratifiers.
- **Tribal & Occupational Health (Pillar 2):** Deploy deep-learning computer vision models for screening Silicosis and Coal Workers' Pneumoconiosis (CWP) via chest X-rays, alongside a probabilistic pedigree network for tracking Sickle Cell Disease in Particularly Vulnerable Tribal Groups (PVTGs).
- **Clinical Decision Support Systems (Pillar 3):** Implement predictive models for High-Risk Pregnancy (HRP) progression, sequential time-series models for Neonatal Sepsis early warnings, and intelligent emergency tele-triage routing. Deploy specialized Vision-Language models to ingest, digitize, and parse unstructured, complex clinical formats

(e.g., handwritten prescriptions, vernacular lab reports) into structured interoperable data formats.

- **Scheme Integrity & Predictive Resource Allocation (Pillar 4):** Establish unsupervised anomaly detection algorithms (Isolation Forests, Graph Neural Networks) to flag upcoding, fraud, and duplicate claims under MASSY and PM-JAY, alongside predictive maintenance models for critical biomedical equipment.
- **Hospital Performance Management (Pillar 5):** real-time healthcare analytics dashboard that merges hospital data into one scorecard to drive smarter management and better patient care.

Vendor shall Leverage state-of-the-art open-weights foundational language models (utilizing Mixture-of-Experts architectures where applicable) that are explicitly pre-trained on Indian languages to ensure high-context processing and culturally nuanced reasoning

4. Integration with Existing Systems

- Build automated, scheduled data extraction and integration pipelines (via RESTful APIs, SFTP, and webhook listeners) with existing state and central platforms.
- Ensure seamless bidirectional data sync with HMIS, e-Shusrut, Nikshay (NTEP), MASSY TMS, RCH Portal, and Sahiya MIS.
- Implement a standardized HL7 FHIR R4 normalization layer to ensure full adherence to Ayushman Bharat Digital Mission (ABDM) data standards.

5. Data Privacy, Governance & Testing

- Institute Privacy by Design, enforcing consent architecture, dynamic de-identification of Personally Identifiable Information (PII) before AI model training, and fine-grained Role-Based Access Control (RBAC).
- Guarantee all AI models act strictly as Clinical Decision Support Systems (CDSS) with a mandatory Human-in-the-Loop (HITL) protocol, ensuring no automated clinical decisions are executed without Medical Officer oversight.
- Conduct comprehensive testing including UAT, algorithmic bias and fairness testing across demographic sub-populations, and CERT-In empanelled Vulnerability Assessment and Penetration Testing (VAPT).

6. Implementation, Rollout & Change Management

- Execute a structured launch starting with a 4-district pilot (Ranchi, East Singhbhum, Khunti, Dumka), followed by an 8-district scale-up, and concluding with 24-district statewide saturation.
- Train ~40,000 Sahiyas via cluster-level hands-on sessions, ~8,000 ANMs via block-level workshops, and ~2,800 Medical Officers via accredited digital CME modules on clinical AI operations.
- Execute a Cascade Training of Trainers (ToT) model.

7. Operations and Maintenance (O&M)

- Provide 24/7 Tier-2 and Tier-3 technical helpdesk operations, bug fixing, and database administration for 48 months post-implementation.
- Establish automated drift detection tracking Population Stability Index (PSI) and execute mandatory quarterly model recalibrations using new clinical datasets.
- Maintain service levels as per agreement and conduct automated, quarterly cloud Disaster Recovery (DR) drills.
- Facilitate and provide data access for independent third-party baseline, mid-term, and end-line econometric impact studies (e.g., Difference-in-Differences methodologies).

8. Grievance Management

- Design, develop, and deploy a centralized, multilingual Grievance Redressal Module accessible across multiple touchpoints, including the Sahiya Mobile App, a public web portal, and a toll-free IVR system. The module must feature automated ticket generation, utilizing natural language processing (NLP) and Speaker Diarization to parse unstructured, code-mixed text and vernacular audio to parse text and vernacular audio complaints, instantly categorizing and routing them (e.g., MASSY/PM-JAY claim denials, drug stock-outs, facility infrastructure issues) to the designated nodal officers.
- Embed the grievance mechanism directly into core healthcare service delivery workflows to ensure transparency and rapid resolution. The system will operate on a strict, time-bound escalation matrix (from Block to District to State levels) driven by predefined Service Level Agreements (SLAs). It must include a closed-loop tracking system that automatically updates beneficiaries and frontline workers on their ticket status via SMS or WhatsApp, ensuring accountability at the point of care.
- Establish an analytical feedback loop that utilizes grievance data as a primary input for systemic healthcare improvement. The platform will deploy trend analytics and anomaly detection on historical grievance logs to identify recurring geographical hotspots, systemic policy bottlenecks, or specific facility-level failures. This intelligence will be fed back to state leadership to refine resource allocation, update clinical and administrative AI predictive models, and adjust frontline workforce training, shifting the state from reactive troubleshooting to proactive service improvement.

9. Documentation and Vendor Exit Transition

- Provide System Requirements Specifications (SRS), technical runbooks, database schemas, and Standard Operating Procedures (SOPs) for data pipeline recovery and security patch deployment.
- Facilitate technical shadowing for state-designated NHM/JAP-IT engineers to build local maintenance capacity.

- Transfer 100% intellectual property, uncompiled source code, Dockerfiles, Kubernetes manifests, trained neural network weight files, and a Zero Encumbrance Certification strictly to the Government of Jharkhand upon exit.

Out of Scope

The following are explicitly excluded from the scope of work:

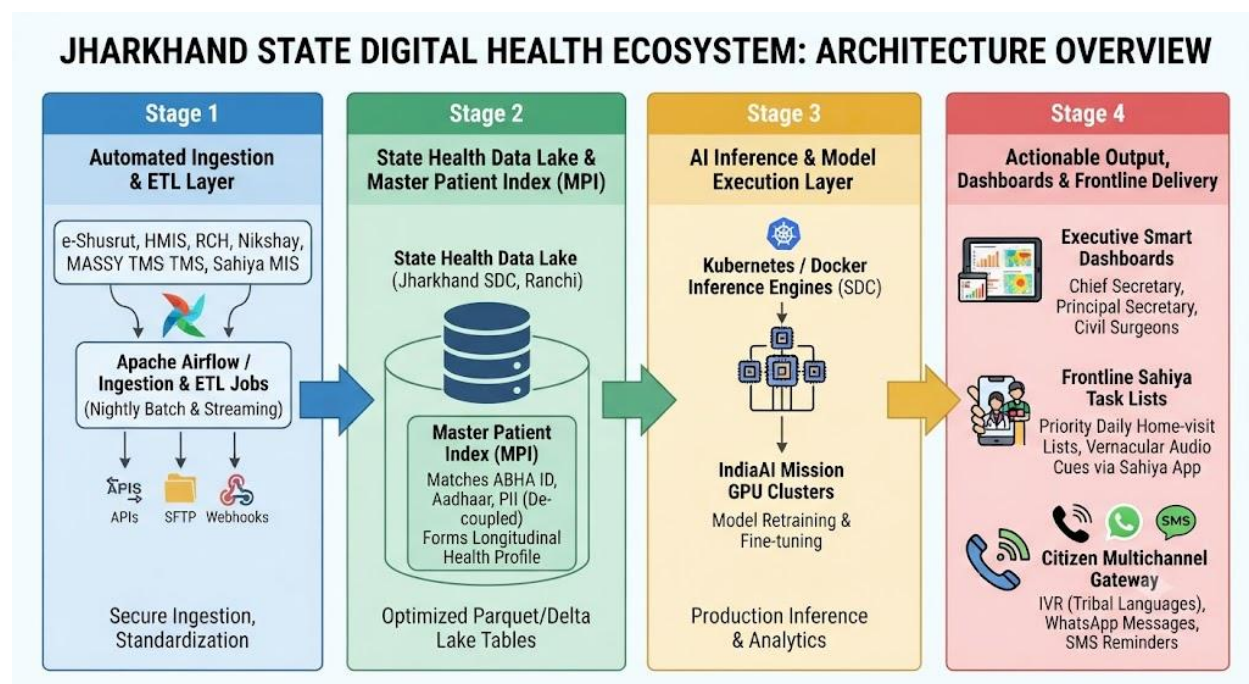
- Costs related to edge compute infrastructure, localized networking, and end-user devices (smartphones, tablets).
- Procurement, installation, or physical deployment of diagnostic equipment (e.g., analog/digital X-ray machines, CT scanners, PACS, or physical DICOM storage arrays).
- Autonomous clinical trials or interventional studies that require separate Central Drugs Standard Control Organisation (CDSCO) or regulatory approvals.

7. Technical Requirements

7.1 System Architecture

The J-HIP platform operates a four-stage automated pipeline that ingests, harmonizes, analyzes, and disseminates health intelligence:

Technical Workflow Stages:



1. Stage 1: Automated Ingestion & ETL Layer:

- Secure RESTful APIs, SFTP batch connectors, and webhook listeners ingest transactional records from state and central platforms (*e-Shusrut, HMIS, RCH, Nikshay, MASSY TMS, and Sahiya MIS*).
- Ingestion jobs are orchestrated using **Apache Airflow**, operating on nightly batch schedules for aggregated indicators and near-real-time streaming for hospital discharge events.

2. Stage 2: State Health Data Lake & Master Patient Index (MPI):

- Ingested data is ingested into the **State Health Data Lake** hosted at the Jharkhand State Data Centre (SDC, Ranchi) formatted in optimized, columnar **Apache Parquet / Delta Lake** tables.
- The **Master Patient Index (MPI)** matches incoming records against the 14-digit **ABHA ID** (Aadhaar-linked Ayushman Bharat Health Account). For records lacking ABHA, deterministic matching (Aadhaar hash, ration card ID) combined with probabilistic demographic matching (name, age, village code, mobile hash) resolves identities into a single longitudinal health profile.
- Personal identifiable information (PII) is decoupled from clinical data using a secure, one-way pseudo-anonymization hashing token.

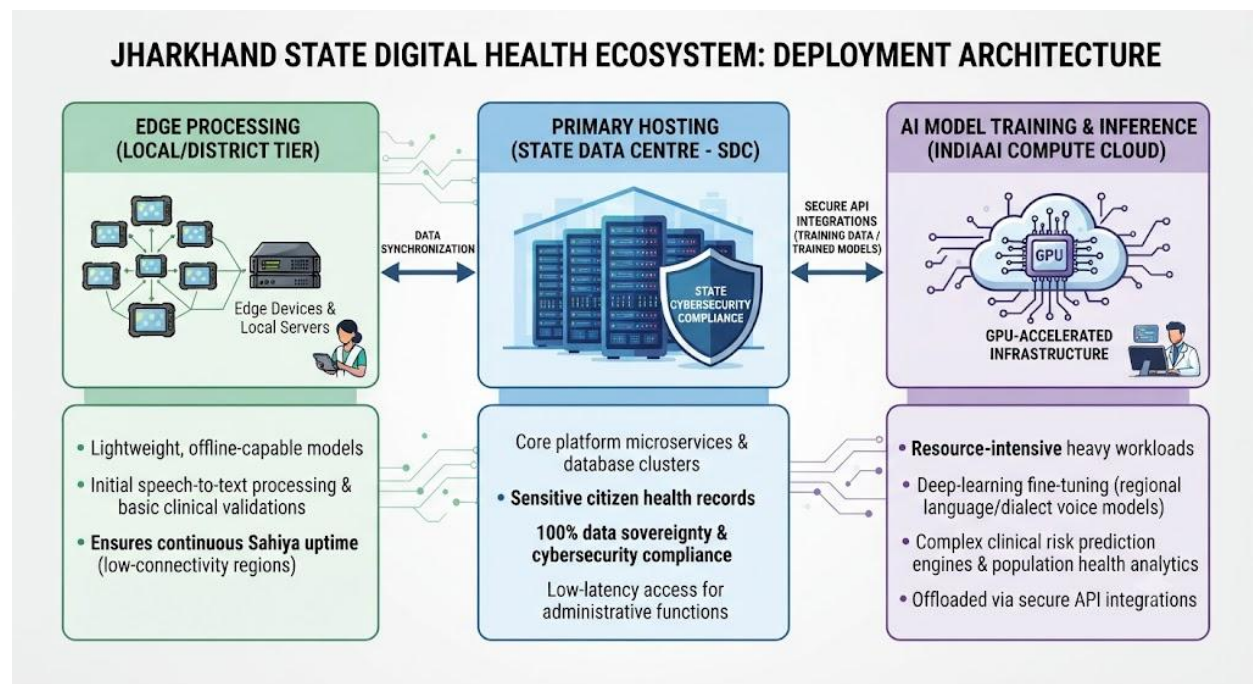
3. Stage 3: AI Inference & Model Execution Layer:

- The standardized feature vectors are passed to production inference engines running in Docker containers orchestrated via Kubernetes at SDC.
- For periodic model retraining, large-scale hyperparameter searches, and speech acoustic model fine-tuning, workloads are dispatched to **IndiaAI Mission subsidized cloud GPU clusters**.

4. Stage 4: Actionable Output, Dashboards & Frontline Delivery:

- Algorithmic outputs are converted into role-based actionable endpoints:
- **Executive Smart Dashboards:** District heat-maps, readmission trends, and DSI indicators for the Chief Secretary, Principal Secretary, and Civil Surgeons.
- **Frontline Sahiya Task Lists:** Priority daily home-visit lists with vernacular audio cues pushed to Sahiya smartphones via the Sahiya App backend API.
- **Citizen Multichannel Gateway:** Automated outbound IVR voice calls in tribal languages (Santhali, Ho, Mundari, Kurukh, Hindi), interactive WhatsApp messages, and SMS reminders.

7.2 Indicative Infrastructure Blueprint



J-HIP adopts a capital-light, hybrid computational architecture that maximizes existing state assets while leveraging national AI infrastructure:

- Primary Hosting (State Data Centre - SDC):** Core platform microservices, database clusters, and sensitive citizen health records are hosted locally within State Data Centre (SDC). This ensures 100% data sovereignty, full compliance with state cybersecurity protocols, and low-latency access for administrative functions.
- Edge Processing (Local/District Tier):** Lightweight, offline-capable models handle initial speech-to-text processing and basic clinical validations on edge devices and district-level servers. This guarantees continuous operational uptime for Sahiyas during field visits, even in low- or zero-connectivity regions.
- AI Model Training & Inference (IndiaAI Compute Cloud):** Resource-intensive heavy workloads—including deep-learning fine-tuning for regional language/dialect voice models, complex clinical risk prediction engines, and large-scale population health analytics—are offloaded via secure API integrations to GPU-accelerated infrastructure provided by the national IndiaAI Mission.

7.3 Data Privacy, Security & DPDP Act 2023 Compliance

As a platform processing sensitive personal health information of ~4.06 Crore citizens, J-HIP embeds **Privacy by Design** and complies with the **Digital Personal Data Protection (DPDP) Act 2023**:

1. Lawful Basis & Explicit Consent Architecture:

- Processing is anchored under lawful public health mandates and explicit electronic consent captured via ABDM Consent Manager protocols. Beneficiaries retain the right to revoke consent for elective follow-up nudges at any time.

2. De-Identification & Data Minimization:

- All data utilized for training AI models or displayed on aggregated administrative dashboards is dynamically scrubbed of direct identifiers (Name, Mobile Number, Aadhaar Number, Exact Village Address).
- Strict purpose limitation ensures health data is never cross-utilized for commercial profiling, credit scoring, or third-party marketing.

3. End-to-End Security:

- **Data at Rest:** Encrypted using **AES-256** standards across all database volumes, data lake partitions, and backup media.
- **Data in Transit:** Enforced **TLS 1.3** transport encryption across all internal microservices and external API endpoints.

4. Immutable Audit Trails & Access Governance:

- Role-Based Access Control (RBAC) ensures users (Doctors, Sahiyas, Civil Surgeons) access only the minimum data necessary for their clinical or administrative role.
- Immutable, tamper-evident audit logs capture every record view, modification, AI inference query, and data export.

5. Mandatory Third-Party Audits:

- Bi-annual Vulnerability Assessment and Penetration Testing (VAPT) conducted by **CERT-In empanelled security auditing agencies**.
- Annual independent DPDP Act privacy compliance and algorithmic bias audits.

7.4 Clinical Safety, Bias Testing & Human-in-the-Loop (HITL) Protocols

The deployment of Artificial Intelligence in healthcare carries profound clinical, ethical, and legal responsibilities. J-HIP enforces strict clinical safety guardrails:

Core Clinical Governance Guardrails:

1. Strictly Advisory Role (Zero Autonomous Medical Decisions):

- All AI-generated outputs-including maternal progression scores, neonatal sepsis warnings, chest X-ray annotations, and claims anomaly flags-are strictly designated as **Clinical Decision Support Advisories**.
- No prescription, medical referral, hospital admission, claim rejection, or punitive action is executed autonomously by the algorithm. Final authority and legal liability remain exclusively with the licensed healthcare professional or authorized government officer.

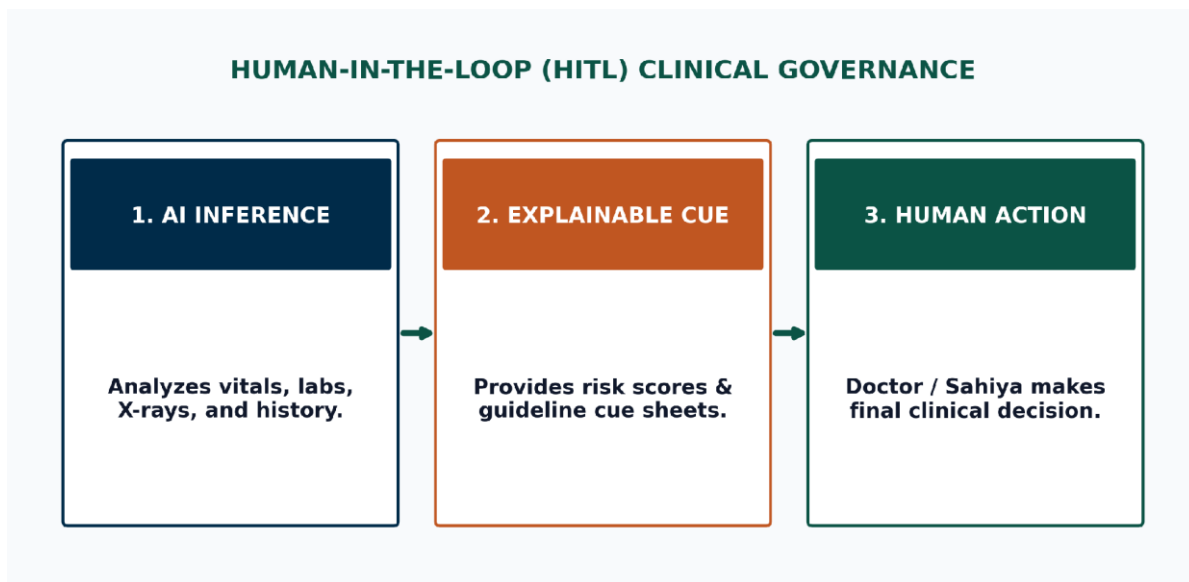


Figure 7.4: Human-in-the-Loop (HITL) Clinical Decision Support Governance Model

2. Explainable AI (XAI) & Interpretability:

- For all clinical risk predictions and claims flags, the platform provides explainable feature contributions using SHAP (Shapley Additive exPlanations) or LIME frameworks. A doctor viewing a High-Risk Pregnancy alert is shown the exact physiological parameters (e.g. *Hb drop of 2.1 g/dl + diastolic BP surge to 95 mmHg + previous history of pre-eclampsia*) that triggered the high-risk classification.

3. Rigorous Algorithmic Bias & Fairness Testing:

- AI models are evaluated across demographic sub-populations to prevent systemic bias:
- Disparate impact ratio and equal opportunity difference metrics are computed across gender, Scheduled Tribe vs. general categories, and remote forest vs. urban blocks.
- Any model exhibiting a predictive performance disparity exceeding 5% across protected groups is automatically flagged for recalibration.

4. Clinical Safety Review Board & Override Logging:

- A multidisciplinary Clinical AI Safety Board comprising senior faculty from RIMS, medical specialists, and public health epidemiologists reviews model accuracy monthly.
- When a clinician overrides an AI recommendation (e.g. classifying a flagged chest X-ray as normal), the clinical rationale is captured in an override log, feeding the continuous MLOps retraining pipeline.

7.5 Independent Third-Party Evaluation Plan

To guarantee scientific rigor, objective validation, and unbiased assessment of public health impact, an independent academic / public health evaluation agency (such as *AIIMS*, *Indian Institute of Public Health - IIPH*, or *National Institute of Health & Family Welfare - NIHF*) will be commissioned:

Evaluation Milestones & Methodology:

1. Baseline Evaluation Study (Months 2–4):

- Rigorous empirical documentation of pre-implementation baselines across pilot and control districts: 30-day readmission rates, baseline maternal complication rates, NCD medication adherence, Sahiya visit frequency, and baseline household out-of-pocket health spending.

2. Mid-Term Evaluation Study (Months 17–19):

- Quasi-experimental evaluation comparing the 4 pilot districts against matched non-pilot control districts using **Difference-in-Differences (DiD)** and **Propensity Score Matching (PSM)** econometric methodologies.
- Assesses preliminary health outcome differentials, frontline worker adoption barriers, and algorithmic accuracy in clinical settings.

3. End-Line Comprehensive Impact Assessment (Months 34–36):

- Comprehensive statewide health outcome evaluation measuring total averted maternal/neonatal deaths, net fiscal savings under MASSY/PM-JAY, out-of-pocket expenditure reductions, and algorithmic fairness across demographic groups.

7.6 Continuous MLOps, Model Drift Detection & Retraining Protocols

Machine learning models deployed in dynamic healthcare environments are subject to **data drift** (shifts in population demographics, seasonal epidemiology, or diagnostic recording habits) and **concept drift** (changes in clinical protocols or disease mutations). J-HIP implements automated MLOps governance:

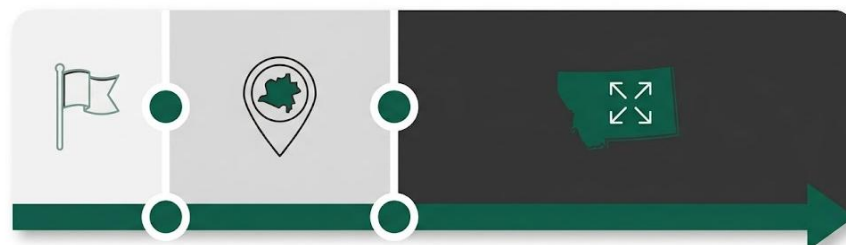
MLOps Governance Rules:

- **Automated Drift Thresholds:** If the Population Stability Index (PSI) for any clinical model exceeds 0.20 or the AUC-ROC drops by > 0.04 against the validation baseline, an automated recalibration alert is triggered.
- **Scheduled Quarterly Retraining:** All risk stratification regressors and NLP acoustic models are retrained every 90 days using the latest anonymized clinical datasets ingested into the State Health Data Lake, leveraging IndiaAI Mission subsidized GPU compute.
- **Zero-Downtime Blue-Green Deployment:** Retrained model versions are deployed using blue-green containerized deployment pipelines with automated rollback capabilities in the event of anomalous prediction variances.

8. Implementation Strategy

8.1 Rollout Roadmap

The Jharkhand Health Intelligence Platform (J-HIP) will be deployed across all **24 districts** of the state. To ensure risk-controlled deployment, continuous operational tuning, and rigorous validation of clinical safety protocols, implementation follows a **two-phase geographic rollout**:



Phase 1

In the pilot phase (month 3-12) the module will be rolled out in 4 representative districts with continued testing and improvements. The SI should ensure approval process of the successful solution during months 9-12:

- **Ranchi:** State capital, tertiary anchor (RIMS), urban-rural mix.
- **East Singhbhum:** Heavy mining/industrial belt (Silicosis hub).
- **Khunti:** Schedule V tribal district (Sickle cell & Mundari voice).
- **Dumka:** Santhal Pargana tribal cluster (Santhali NLP & HRP).

Phase 2

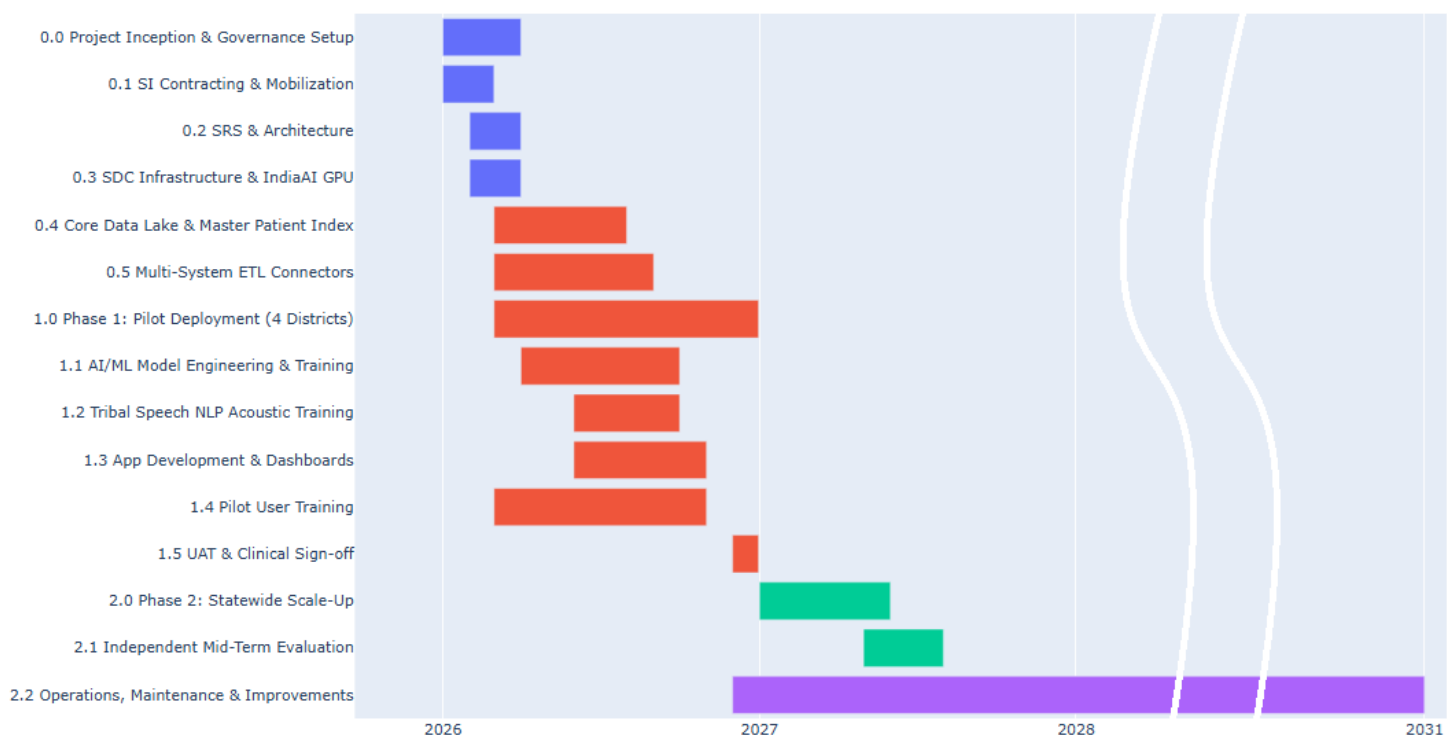
In the scale-up phase (month 12 onwards), SI will roll out the successful solution to all districts of Jharkhand.

Project Phase	Timeline	Core Workstreams & Key Deliverables
Phase 0: Inception & Governance	Months 1–3	1. Notification of State Apex Steering Committee & SPMU constitution. 2. Onboarding of System Integrator & technical kick-off. 3. Finalization of System Requirements Specifications (SRS) & FHIR mapping. 4. Provisioning of SDC server environments & IndiaAI compute access. 5. Inter-departmental data-sharing agreements (Nikshay, MASSY, e-Shusrut). 6. Development of State Health Data Lake & Master Patient Index (MPI). 7. Integration of ETL pipelines (HMIS, RCH, Nikshay, MASSY, Sahiya)
Phase 1: Build & 4-District Pilot	Months 4–12	1. Development & lab validation of AI models (Pillars 1–4). 2. Training of Trainers (ToT) & training in pilot districts. 3. Field validation and Deployment of Solution in 4 districts 4. Algorithmic recalibration based on results 5. Onboarding district users 6. Independent Evaluation for Statewide Rollout
Phase 2: Statewide Rollout	Months 12 onwards	1. Expansion to remaining districts 2. Full state-wide activation of all 27 AI use cases. 3. Executive Dashboard launch for Chief Secretary & Cabinet. 4. 24/7 Operations & Maintenance (O&M) tier-2/3 technical support. 5. Continuous MLOps, model drift recalibration & quarterly security audits. 6. Annual refresher training for frontline cadres.

Detailed Plan

WBS Code	Workstream / Task Description	Lead Responsibility	Start Month	End Month
0.0	Project Inception & Governance Setup	DoHMEFW	Month 1	Month 3
0.1	System Integrator Contracting & Mobilization	Implementation Agency	Month 1	Month 2
0.2	System Requirements Specification (SRS) & Architecture	SI Enterprise Architect	Month 2	Month 3
0.3	SDC Infrastructure & IndiaAI GPU Allocation	State IT / MeitY / SI	Month 2	Month 3
0.4	Core Data Lake & Master Patient Index (MPI)	SI Data Team	Month 3	Month 7
0.5	Multi-System ETL Connectors (HMIS, RCH, Nikshay)	SI ETL Developers	Month 3	Month 8
1.0	Phase 1: Pilot Deployment (4 Districts)	SI / Pilot DHS Teams	Month 3 onwards	Month 12
1.1	AI/ML Model Engineering & Algorithm Training	SI AI/ML Lead	Month 4	Month 9
1.2	Tribal Speech NLP Acoustic Training (Santhali/Ho)	SI NLP Specialists	Month 6	Month 9

WBS Code	Workstream / Task Description	Lead Responsibility	Start Month	End Month
1.3	Application Development & Dashboards	SI Dev Team	Month 6	Month 10
1.4	Pilot User Training (~7,000 Sahiyas & 500 Doctors)	SI Training Lead	Month 3 onwards	Month 10
1.5	Pilot User Acceptance Testing (UAT) & Clinical Sign-off	Clinical Review Board	Month 12	Month 12
2.0	Phase 2: Statewide Scale-Up	SI / DHS Teams	Month 13	Month 18
2.1	Independent Mid-Term Evaluation Study	3rd-Party Agency	Month 17	Month 19
2.2	Operations, maintainance and continues improvements	SI / State Digital Cell	Month 12	Month 60



8.2 Implementation Modality

The implementation of the Jharkhand Health Intelligence Platform (J-HIP) will follow a System Integrator (SI)-led implementation model under the administrative and technical oversight of the Department of Health, Medical Education & Family Welfare (DoHMEFW) and the State Health Society (National Health Mission), Jharkhand.

The System Integrator will be selected through an open, transparent competitive bidding process adhering to the Jharkhand Procurement Policy and CVC guidelines.

Milestone-Linked Capital Payments:

- Implementation CapEx payments to the System Integrator will be strictly linked to verified, tangible deliverables and user-accepted milestones:

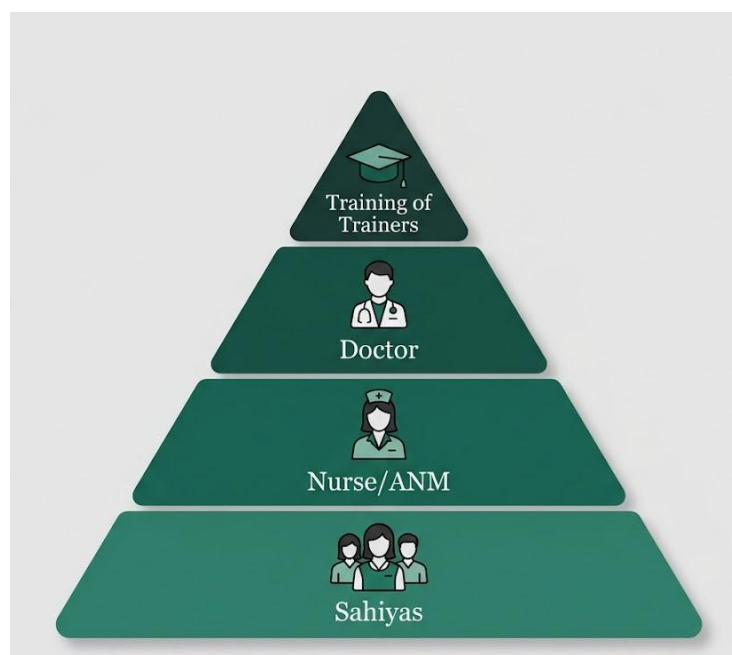
Milestone	Phase / Description	Payment	Deliverable / Condition
1	Mobilization & Inception Approval	10%	Against approved Work Breakdown Structure and bank guarantee.
2	Core Platform & Data Lake Go-Live at SDC	20%	Upon verified SDC setup, MPI engine, and FHIR ETL pipelines.
3	AI Engines & Use Cases Integration	25%	Upon completed development and UAT of Pillars 1–4 modules.
4	Successful 4-District Pilot Acceptance	20%	Upon field validation in Ranchi, East Singhbhum, Khunti, and Dumka.
5	Statewide 24-District Go-Live	25%	Upon successful statewide rollout and departmental sign-off.

SLA-Linked Quarterly Operational Payments:

- The 48-month Operations & Maintenance (O&M) phase will be disbursed in equal quarterly installments strictly tied to Service Level Agreement (SLA) performance (uptime $\geq 99.5\%$, incident resolution times, model drift monitoring, helpdesk responsiveness). Please refer *Annexure A* for suggested SLA.

8.3 Health Workforce Capacity Building & Training Plan

The success of J-HIP hinges on user adoption across Jharkhand's diverse public health workforce. The capacity building program employs a **Cascade Training of Trainers (ToT) model**:



Structured Training Modules by Cadre

Target Cadre	Total Numbers	Training Modality & Duration	Core Curriculum Focus	Training Material Format
Frontline Sahiyas & Sahiya Sathis	~40,000	1-Day Cluster-level practical hands-on session + Monthly refresher at VHSNC	1. Navigating AI-prioritized daily task lists on Sahiya App. 2. Interpreting post-discharge red-flag checklists. 3. Conducting vernacular follow-up visits in Santhali/Ho. 4. Escalating emergency cases to 108/MOIC.	Illustrated pictorial handbook in Hindi/tribal scripts + Vernacular video tutorials on mobile
ANMs, CHOs & Staff Nurses	~8,000	2-Day Block-level workshop with live case simulations	1. Point-of-care vital sign and lab anomaly data capture. 2. Maternal HRP progression alert interpretation. 3. SNCU/NBSU neonatal sepsis early warning triaging. 4. Sahiya task supervision and escalation closure.	Digital interactive SOP modules + Video case walkthroughs
Medical Officers & Specialists	~2,800	1-Day Clinical CME module at District Hospital Hubs	1. Clinical Decision Support System (CDSS) operational bounds. 2. Chest X-ray AI reading for Silicosis, CWP, and TB. 3. Human-in-the-Loop clinical override protocols and liability. 4. Intelligent referral routing via 104/tele-triage.	Accredited CME digital modules + Clinical algorithm quick-reference cards

Target Cadre	Total Numbers	Training Modality & Duration	Core Curriculum Focus	Training Material Format
District Leadership & DDMs	~150	1-Day Executive briefing at State / Zonal HQ	1. Executive Dashboard navigation & DSI interpretation. 2. Monitoring avoidable readmission and disease heat-maps. 3. Data pipeline monitoring and data quality correction workflows. 4. Reviewing claims anomaly flags with SHA auditors.	Executive Dashboard User Guide + Video analytics tutorials

9. Financial Details

This schedule outlines the comprehensive financial model for the Jharkhand Health Intelligence Platform across implementation and sustained multi-year operations, establishing the resource allocations, infrastructure sizing, and budgetary phasing. The cost estimates are based on assumptions stated regarding team size, AI model used and cost per resource provided below. These are tentative estimates and cost discovery may be done through competitive bidding process.

9.1 Master Financial Summary & 5-Year Cash Flow Phasing

Master budgetary distribution across development and operational heads, detailing annual phasing, risk contingencies, and statutory taxes.

S.No	Cost Component / Budget Head	Year 1 (₹ Cr)	Year 2 (₹ Cr)	Year 3 (₹ Cr)	Year 4 (₹ Cr)	Year 5 (₹ Cr)	Total 5-Yr (₹ Cr)	% Share
1	SI Technical Development Manpower (NICS Based)	15.69	0.00	0.00	0.00	0.00	15.69	20.34%
2	O&M Support & Helpdesk Manpower (48 Months)	0.00	3.65	3.65	3.65	3.65	14.60	18.92%
3	Core Platform Engines — Dev Manpower (NICS Based)	8.50	0.00	0.00	0.00	0.00	8.50	11.02%
4	Core Platform Engines — O&M Manpower (NICS Based, 48m)	0.00	1.56	1.56	1.56	1.56	6.23	8.08%
5	IndiaAI GPU Infrastructure — Dev Training (H200 80GB)	0.15	0.00	0.00	0.00	0.00	0.15	0.19%
6	IndiaAI GPU Infrastructure — O&M Hosting & Retraining (48m)	0.00	0.09	0.09	0.09	0.09	0.34	0.45%
7	LLM Token Inference — IndiaAI Managed Server Pricing (48m)	0.00	0.10	0.17	0.17	0.17	0.62	0.80%
8	State Data Centre CPU Infrastructure (Govt SDC — Zero Cost)	0.00	0.00	0.00	0.00	0.00	0.00	0.00%
9	Operational Overheads (Phased across 5 Years)	5.47	1.31	1.31	1.31	1.31	10.72	13.89%
—	SUBTOTAL: BASE PROGRAM ESTIMATE	29.82	6.71	6.78	6.78	6.78	56.85	73.69%
16	Contingency & Risk Buffer (@ 10% of Base)	2.98	0.67	0.68	0.68	0.68	5.68	7.37%

Detailed Project Report - Jharkhand Health Intelligence Platform (J-HIP)

17	Executing Agency Overhead (@ 5% of Base)	1.49	0.34	0.34	0.34	0.34	2.84	3.68%
18	Applicable Statutory GST (@ 18% on Services)	6.17	1.39	1.40	1.40	1.40	11.77	15.25%
—	GRAND TOTAL BUDGET ENVELOPE (ALL INCLUSIVE)	40.46	9.10	9.19	9.19	9.19	77.15	100.00%

Key Methodology & Notes:

- Year 1 encompasses implementation, core data pipeline build, custom algorithm training, and district pilots.
- Years 2 to 5 provide sustained statewide operational support, helpdesk operations, continuous model retraining, and field change management across 24 districts.
- Statutory GST and administrative overheads are calculated based on applicable government procurement guidelines.

9.2 Use Case Machine Learning Taxonomy & Sizing Matrix

Functional classification of all platform modules across disease domains, data modalities, development compute requirements, and operational hosting architectures.

ID	Use Case Module	Pillar	Target ML Algorithm / Framework	Input Modalities & Hosting	Dev GPU (Hrs)	O&M GPU (Hrs)
1.1	Post-discharge care & complication-risk management	Pillar 1	Gradient Boosted Decision Trees (XGBoost/CatBoost)	EHR, Discharge Records, Vitals, Comorbidities <i>[Enterprise CPU (SDC) real-time scoring]</i>	250	0
1.2	Chronic-disease care-gap & multimorbidity management	Pillar 1	Longitudinal Recurrent Neural Networks (LSTM) + LightGBM	Longitudinal Visits, Refill Logs, Lab Trends <i>[Enterprise CPU (SDC) nightly batch scoring]</i>	300	0
1.3	Maternal, postnatal & newborn continuity tracking	Pillar 1	Time-Series Forecasting + State Machine Engine	ANC/PNC Registration Sequence, Session Logs <i>[Enterprise CPU (SDC) daily queueing]</i>	0	0
1.4	SAM/MTC discharge & nutritional-relapse risk	Pillar 1	Random Forest Classifier + Growth Curve Analytics	Anthropometric Trajectories, Weight Curves, MTC Logs <i>[Enterprise CPU (SDC) growth tracking]</i>	0	0
1.5	Immunisation defaulter & dropout recovery	Pillar 1	Probabilistic Dropout Scoring (LightGBM)	U-WIN Vaccination Records, Child Age, Geography <i>[Enterprise CPU (SDC) due list generation]</i>	0	0
1.6	TB treatment-adherence & follow-up tracking	Pillar 1	Time-Series Classifier + Nikshay Linkage Engine	Nikshay Refill Logs, Treatment Supporter Notes <i>[Enterprise CPU (SDC) adherence risk flags]</i>	0	0
1.7	AI-guided frontline visit prioritisation	Pillar 1	Multi-Objective Constraint Optimization (Linear Program + XGBoost)	Combined Approved Risk Flags, Workload Limits, Geo-location <i>[Enterprise CPU (SDC)]</i>	200	600

ID	Use Case Module	Pillar	Target ML Algorithm / Framework	Input Modalities & Hosting	Dev GPU (Hrs)	O&M GPU (Hrs)
				<i>tasking + GPU Speech Synthesis]</i>		
2.1	Occupational respiratory-risk & longitudinal follow-up	Pillar 2	Deep CNN (ResNet-50 / DenseNet-121) + XGBoost	Chest X-Rays (DICOM/JPEG), Spirometry, Mining Logs <i>[1x H200 GPU (IndiaAI) on-demand inference]</i>	400	1200
2.2	Silicosis/CWP-TB coordinated care & Nikshay linkage	Pillar 2	Multi-Modal Vision Transformer (ViT) + Graph Linkage	Chest X-Rays, Nikshay TB Events, Clinical Logs <i>[1x H200 GPU (IndiaAI) dual-screening inference]</i>	400	1200
2.3	Sickle-cell & anaemia screening-to-care completion	Pillar 2	Decision Trees + Referral Tracking Rules Engine	Screening Results, HPLC Confirmatory Tests, Care Logs <i>[Enterprise CPU (SDC) referral monitoring]</i>	0	0
3.1	High-risk pregnancy progression & escalation support	Pillar 3	Explainable XGBoost (SHAP/LIME) + Escalation Matrix	Obstetric Parameters, Vitals, Ultrasound Metadata <i>[Enterprise CPU (SDC) real-time CDSS engine]</i>	350	0
3.2	Neonatal deterioration & sepsis early-warning support	Pillar 3	Convolutional 1D / LSTM Neural Networks on Vitals	Continuous Vital Sign Trends, SNCU/NBSU Logs <i>[Enterprise CPU (SDC) vital stream scoring]</i>	350	0
3.3	AI-assisted longitudinal patient summary	Pillar 3	LLM (Llama-3 8B / BioBERT) NLP Summarization Engine	Multi-source EHR, ABHA Records, E-Sushrut Notes <i>[Token-based / GPU micro-instance generation]</i>	450	1000
3.4	Medication-safety & guideline-concordance support	Pillar 3	Knowledge Graph + Rule-Based Decision Support System	Prescriptions, Drug Allergies, Diagnoses, Guidelines <i>[Enterprise CPU (SDC) drug interaction checks]</i>	0	0
3.5	Inpatient deterioration & discharge-readiness support	Pillar 3	Modified Early Warning Score (MEWS) XGBoost Classifier	Inpatient Observation Charts, Lab Trends, Comorbidities <i>[Enterprise CPU (SDC) ward monitoring dashboard]</i>	0	0
3.6	Closed-loop referral outcome tracking	Pillar 3	Graph Traversal Algorithm + State Machine Engine	Referral Events, Facility Arrival, Admission Logs <i>[Enterprise CPU (SDC) referral completion tracking]</i>	0	0
4.1	Executive Financial Command & Multi-Grant Reconciliation	Pillar 4	Multi-Source Financial ETL + Graph Entity Resolution	Kuber IFMS, PFMS/SNA, DMFT, XV-FC Feeds <i>[Enterprise CPU (SDC) real-time master ledger]</i>	0	0
4.2	MASSY & PM-JAY Claims Anomaly Detection	Pillar 4	Graph Neural Networks (GNN) + Isolation Forests	Claims Data, Package Combos, Provider Network Graphs	400	400

ID	Use Case Module	Pillar	Target ML Algorithm / Framework	Input Modalities & Hosting	Dev GPU (Hrs)	O&M GPU (Hrs)
				<i>[Enterprise CPU / GPU batch audit scoring]</i>		
4.3	Scheme Entitlement & Coverage Gap Identification	Pillar 4	Rule-Based Expert System + Ration Card / SECC Linkage	Socio-economic Data, PM-JAY Enrolment, Ration Registers <i>[Enterprise CPU (SDC) eligibility matching]</i>	0	0
4.4	PC&PNDT Compliance Risk & Inspection Prioritization	Pillar 4	Statistical Anomaly Detection + Form-F Risk Scoring	Clinic Reporting Logs, Form F Registrations, Garima Records <i>[Enterprise CPU (SDC) inspection prioritisation]</i>	0	0
4.5	Pre-Audit & AC/DC Bill Compliance Agent	Pillar 4	Document OCR + JFR Treasury Rule Verification Engine	Vendor Vouchers, Subvouchers, GSTIN Entries, JFR Rules <i>[Enterprise CPU (SDC) pre-audit checklist assistant]</i>	250	200
4.6	Multi-Grant Fund Velocity & UC Trigger Engine	Pillar 4	Time-Series Velocity Analytics + Auto-Draft UC Rules	Treasury Disbursement Logs, SNA Balances, Utilization Targets <i>[Enterprise CPU (SDC) UC milestone alerts]</i>	0	0
5.1	Facility-wise patient volume and speciality tracking	Pillar 5	Encounter Classification + Outlier/Zero-Reporting Engine	HMIS Footfall, e-Shusrut Encounter Logs, Speciality Master <i>[Enterprise CPU (SDC) activity dashboard]</i>	0	0
5.2	Biomedical equipment utilisation & non-reporting flagging	Pillar 5	Asset Usage Cross-Reconciliation & Anomaly Detection	Biomedical Asset Register, Report Logs, Test Counts <i>[Enterprise CPU (SDC) non-functional asset alert]</i>	0	0
5.3	Hospital workforce (HR) availability & deployment tracking	Pillar 5	HRIS Roster Matching + Vacancy Mismatch Analytics	HRIS Attendance Data, Sanctioned Posts, Department Rosters <i>[Enterprise CPU (SDC) vacancy & mismatch monitor]</i>	0	0
5.4	Facility and scheme financial-flow tracking	Pillar 5	Budget vs Expenditure Velocity & Discrepancy Flagging	Facility Budget Allocations, NHM PIP Heads, Claims Data <i>[Enterprise CPU (SDC) facility fiscal monitor]</i>	0	0
5.5	Integrated facility performance scorecard	Pillar 5	Multi-Criteria Composite Index + Generative Narrative Engine	5.1-5.4 Outputs, HMIS Quality Indicators, Benchmarks <i>[Enterprise CPU (SDC) + GPU NLP scorecard generator]</i>	250	400
Shared-Voice	Mandatory Vernacular Voice Support (Santhali/Ho/Kurukh/Mundari)	Shared Core	Conformer / Whisper ASR & Tacotron2 / VITS TTS Models	Audio Recordings, Phonetic Dictionaries in Tribal Languages	450	1200

ID	Use Case Module	Pillar	Target ML Algorithm / Framework	Input Modalities & Hosting	Dev GPU (Hrs)	O&M GPU (Hrs)
				[1x H200 GPU (IndiaAI) continuous voice hosting]		
TOTAL	TOTAL 27 USE CASES + SHARED VERNACULAR VOICE MODULES				4050	6200

Taxonomy Summary:

- Tabular risk prediction models run efficiently on enterprise CPU infrastructure at the State Data Centre.
- Deep learning computer vision (X-Ray screening) and tribal speech models utilize IndiaAI subsidized GPU compute.
- Clinical summarization and automated audit narratives leverage self-hosted generative models via managed token endpoints.

9.3 System Integrator Development Team Manpower (12 Months)

Resource staffing matrix and technical profile allocations for the system integrator development team across the initial implementation phase.

S.No	Role / Technical Profile	NICSI Code	Experience Band	Qty	Months	Total MM	Total Dev Cost (₹)
1	Engagement Lead / Director	R1	Consultant: 15 years and above	1	6	6	₹2,782,292.58
2	Project Manager / Scrum Master	R2	Consultant: 10 years to <15 years	2	12	24	₹9,539,296.32
3	Lead Healthcare Architect	R1	Consultant: 15 years and above	1	12	12	₹5,564,585.16
4	Senior AI/ML Solution Architect	R2	Consultant: 10 years to <15 years	1	12	12	₹4,769,648.16
5	Senior AI/ML & NLP Engineers	R3	Consultant: 6 years to <10 years	4	12	48	₹17,488,692.48
6	Lead Data Engineers & Lake Architects	R3	Consultant: 6 years to <10 years	3	12	36	₹13,116,519.36
7	Full-Stack Developers (Web/API)	R4	Consultant: 3 years to <6 years	6	12	72	₹22,417,341.12
8	Mobile App & Workflow Engineers	R4	Consultant: 3 years to <6 years	3	12	36	₹11,208,670.56
9	DevOps & Cloud Security Engineers	R3	Consultant: 6 years to <10 years	2	12	24	₹8,744,346.24
10	Medical Informatics & Clinical Specialist	R6	Subject Matter Expert: 10 years to <15 years	2	6	12	₹6,359,535.24
11	QA, Test & Security Engineers	R5	Consultant: <3 years	3	12	36	₹10,016,245.44
12	Training & Change Management Leads	R4	Consultant: 3 years to <6 years	4	9	36	₹11,208,670.56
13	Technical Documentation Specialists	R5	Consultant: <3 years	2	6	12	₹3,338,748.48
14	Clinical Data Analyst	R4	Consultant: 3 years to <6 years	2	12	24	₹7,472,447.04
15	NLP & Speech Processing Engineer	R3	Consultant: 6 years to <10 years	2	10	20	₹7,286,955.20
16	Healthcare Domain Expert / AYUSH-NDHM Advisor	R6	Subject Matter Expert: 10 years to <15 years	1	12	12	₹6,359,535.24
17	DPDP & Data Privacy Compliance Specialist	R3	Consultant: 6 years to <10 years	1	6	6	₹2,186,086.56

Detailed Project Report - Jharkhand Health Intelligence Platform (J-HIP)

18	Business Analyst & Requirements Lead	R4	Consultant: 3 years to <6 years	1	12	12	₹3,736,223.52
19	Data Annotation & Labeling Specialist	R5	Consultant: <3 years	2	6	12	₹3,338,748.48
TOTAL	TOTAL SI IMPLEMENTATION MANPOWER			43	—	452	₹156,934,587.74

Manpower Notes:

- Monthly resource tariffs are derived from the official NICSI empanelled consultancy rate schedule.
- Loaded rates include mandatory statutory allowances and operating margins, excluding GST.

9.4 Operations & Maintenance (O&M) Support Team (48 Months)

Sustained engineering, data operations, clinical audit support, and helpdesk staffing deployed across the multi-year operational period.

S.No	Role / Technical Profile	NICSI Code	Experience Band	Qty	Months	Total MM	Total 48-Mo Cost (₹)
1	Project Manager / O&M Lead	R2	Consultant: 10 years to <15 years	1	48	48	₹19,078,592.64
2	Data & API Engineer	R4	Consultant: 3 years to <6 years	1	48	48	₹14,944,894.08
3	Application Support Engineer (L2/L3)	R5	Consultant: <3 years	2	48	96	₹26,709,987.84
4	DevOps & Infrastructure Lead	R3	Consultant: 6 years to <10 years	1	48	48	₹17,488,692.48
5	QA & Release Engineer	R5	Consultant: <3 years	1	48	48	₹13,354,993.92
6	Clinical Quality & Audit Support Analyst	R6	Subject Matter Expert: 10 years to <15 years	1	24	24	₹12,719,070.48
7	24x7 Helpdesk & Tier-2 Support Engineers	R5	Consultant: <3 years	2	48	96	₹26,709,987.84
8	Dashboard & Reporting Analyst	R4	Consultant: 3 years to <6 years	1	48	48	₹14,944,894.08
TOTAL	TOTAL O&M SUPPORT MANPOWER			10	—	456	₹145,951,113.36

O&M Staffing Notes:

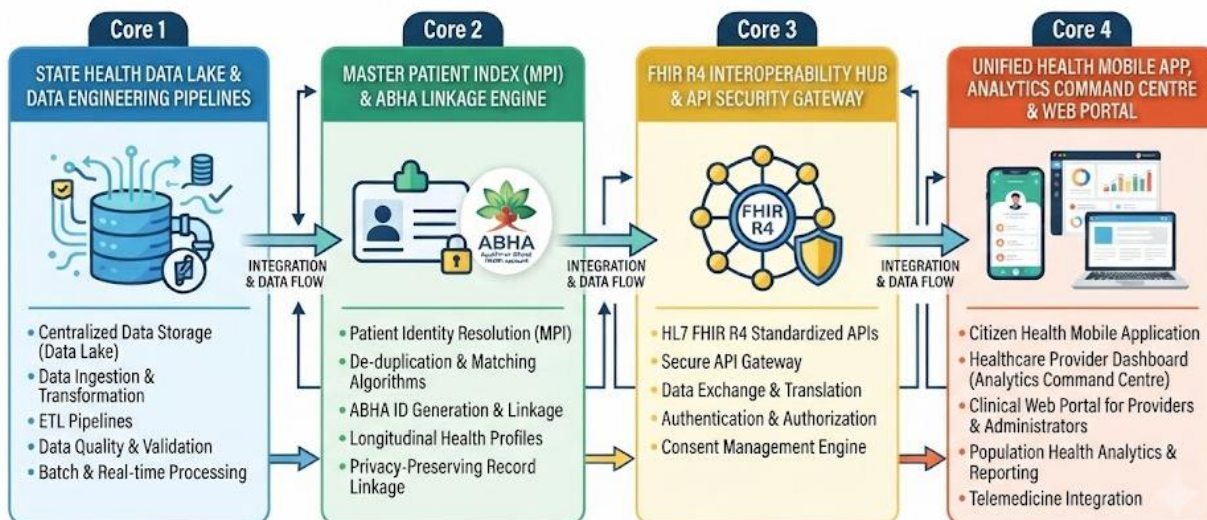
- Ensures continuous 24x7 platform reliability, data sync monitoring, model retraining, and district support throughout the lifecycle.

9.5 Core Platform Architecture Engines Manpower

Dedicated engineering allocations for foundation data infrastructure, Master Patient Index entity resolution, FHIR API gateway, and unified web/mobile portals.

Core Platform Architecture:

- Core 1: State Health Data Lake and data engineering pipelines.
- Core 2: Master Patient Index (MPI) and ABHA linkage engine.
- Core 3: Clinically compliant FHIR R4 interoperability hub and API security gateway.
- Core 4: Unified health mobile application, analytics command centre, and web portal.



S.No	Engine	Role / Technical Profile	Cod e	Qt y	Mos	Tota l MM	Rate / Mo (₹)	Total Cost (₹)
A. CORE PLATFORM DEVELOPMENT TEAM (12-MONTH IMPLEMENTATION)								
1	Core 1	Data Lake Architect	R2	1	12	12	₹397,470.68	₹4,769,648.16
2	Core 1	Senior Data Engineers	R3	3	12	36	₹364,347.76	₹13,116,519.36
3	Core 1	MLOps & Model Registry Engineer	R3	1	10	10	₹364,347.76	₹3,643,477.60
4	Core 1	DevOps / Kubernetes Engineer	R4	1	10	10	₹311,351.96	₹3,113,519.60
5	Core 2	Solution Architect (Identity & Resolution)	R2	1	8	8	₹397,470.68	₹3,179,765.44
6	Core 2	Backend Developers (Entity Resolution)	R4	2	12	24	₹311,351.96	₹7,472,447.04
7	Core 2	Database & Search Specialist (PG/ES)	R3	1	8	8	₹364,347.76	₹2,914,782.08
8	Core 2	ABHA Integration Developer	R4	1	8	8	₹311,351.96	₹2,490,815.68
9	Core 3	Integration Architect (HL7/FHIR)	R2	1	8	8	₹397,470.68	₹3,179,765.44
10	Core 3	API Gateway Developers	R4	2	10	20	₹311,351.96	₹6,227,039.20
11	Core 3	FHIR Resource Specialist	R3	1	8	8	₹364,347.76	₹2,914,782.08
12	Core 3	API Security & Gateway Developer	R3	1	8	8	₹364,347.76	₹2,914,782.08
13	Core 4	UI/UX Lead & Design System Architect	R3	1	12	12	₹364,347.76	₹4,372,173.12
14	Core 4	Mobile Application Developers (Flutter/Android)	R4	3	12	36	₹311,351.96	₹11,208,670.56
15	Core 4	Frontend Dashboard Developers (React/D3.js)	R4	2	12	24	₹311,351.96	₹7,472,447.04
16	Core 4	Voice UI & Accessibility Engineer	R3	1	8	8	₹364,347.76	₹2,914,782.08
17	Core 4	UX / Accessibility Designer	R4	1	10	10	₹311,351.96	₹3,113,519.60
—	SUBTOTAL CORE PLATFORM DEVELOPMENT MANPOWER					250	—	₹85,018,936.16
B. CORE PLATFORM O&M SUPPORT TEAM (48 MONTHS SUSTAINED OPERATIONS)								
1	Core 1	Lead Data Engineer / Lake Specialist	R3	1	48	48	₹364,347.76	₹17,488,692.48

2	Core 2	ABHA/MPI Integration Specialist	R4	1	48	48	₹311,351.96	₹14,944,894.08
3	Core 3	FHIR & API Gateway Engineer	R4	1	48	48	₹311,351.96	₹14,944,894.08
4	Core 4	Unified Mobile App & Dashboard Maintenance	R4	1	48	48	₹311,351.96	₹14,944,894.08
—	SUBTOTAL CORE PLATFORM O&M MANPOWER					192	—	₹62,323,374.72
TOTAL	TOTAL CORE PLATFORM MANPOWER (DEV + O&M)					442	—	₹147,342,310.88

9.6 IndiaAI Mission GPU Cloud Infrastructure

High-performance GPU cluster capacity requirements and subsidized tariff costing for training and continuous hosting workloads.

S.No	AI Workload / Task Description	GPU Model	Compute (Hrs)	Rate / Hr (₹)	Total Cost (₹)
A. DEVELOPMENT TRAINING GPU COMPUTE (12 MONTHS)					
1	Tribal Speech NLP Fine-Tuning (Santhali/Ho ASR/TTS)	NVIDIA H200 80GB (1x)	700	₹375.00	₹262,500.00
2	Computer Vision X-Ray Models (Silicosis/TB)	NVIDIA H200 80GB (1x)	800	₹375.00	₹300,000.00
3	Maternal HRP & Neonatal Clinical Risk Models	NVIDIA H200 80GB (1x)	700	₹375.00	₹262,500.00
4	Claims Anomaly Detection Graph Neural Networks	NVIDIA H200 80GB (1x)	400	₹375.00	₹150,000.00
5	LLM Clinical Summary LoRA Fine-Tuning (Llama-3/BioBERT)	NVIDIA H200 80GB (1x)	900	₹375.00	₹337,500.00
6	Dashboard Analytics & Scorecard NLP Model Training	NVIDIA H200 80GB (1x)	500	₹375.00	₹187,500.00
—	SUBTOTAL DEV TRAINING GPU COMPUTE		4,000	—	₹1,500,000.00
B. O&M HOSTING & MODEL RETRAINING GPU COMPUTE (48 MONTHS)					
1	Vernacular Voice Speech Synthesis Hosting (Triton co-located)	NVIDIA H200 80GB (1x)	2,000	₹375.00	₹750,000.00
2	On-Demand Diagnostic X-Ray Vision Inference	NVIDIA H200 80GB (1x)	2,400	₹375.00	₹900,000.00
3	LLM Patient Summary Token Inference (Production)	NVIDIA H200 80GB (1x)	2,000	₹375.00	₹750,000.00
4	Dashboard Real-Time Analytics NLP Inference	NVIDIA H200 80GB (1x)	1,600	₹375.00	₹600,000.00
5	Continuous Model Drift Recalibration & Retraining	NVIDIA H200 80GB (1x)	1,200	₹375.00	₹450,000.00
—	SUBTOTAL O&M HOSTING GPU COMPUTE		9,200	—	₹3,450,000.00
TOTAL	TOTAL GPU COMPUTE INFRASTRUCTURE		13,200	—	₹4,950,000.00

GPU Compute Basis:

- Tariff rates reflect official subsidized rates under the IndiaAI Mission compute allocation scheme for state digital health implementations.
- These cost will depend on usage and adoption

9.7 Generative LLM & Token Inference

Volume projections and multi-year costing for self-hosted LLM inferences, clinical narratives, audio transcription, and executive insights.

S.No	Service / NLP Pipeline	Model Architecture	Tokens/ Call	Monthly Calls (Y2)	Monthly Calls (Y3-5)	Rate / 1K (₹)	Yr 2 (₹ L)	Yr 3-5 (₹ L/yr)	Total (₹ L)
1	3.3 AI-Assisted Longitudinal Patient Summary	Llama-3 8B (LoRA)	2500	70,000	120,000	0.150	3.15	5.40	19.35
2	3.4 Medication Safety & Guideline Concordance	BioBERT / KG NLP	1000	50,000	80,000	0.100	0.60	0.96	3.48
3	4.1 & 4.2 MASSY/PM-JAY & Grant Reconciliation Audit NLP	Llama-3 8B	1500	30,000	50,000	0.150	0.81	1.35	4.86
4	1.7 Frontline Visit Audio Transcript Summary & Prompts	Whisper ASR + NLP	1200	50,000	80,000	0.120	0.86	1.38	5.01
5	4.4 & 4.5 PC&PNDT & Pre-Audit Bill Compliance Agent NLP	OCR-LayoutLM & Rules	600	5,000	8,000	0.080	0.03	0.05	0.17
6	Shared: Vernacular Voice ASR Tokenization (Santhali/Ho)	Conformer ASR	800	70,000	110,000	0.080	0.54	0.84	3.07
7	5.5 Executive Performance Scorecard & Real-Time Analytics NLP	Llama-3 8B	1500	110,000	180,000	0.120	2.38	3.89	14.04
8	1.2 Chronic-Disease Population Health Scoring Summary	BioBERT / NER	1000	80,000	120,000	0.100	0.96	1.44	5.28
9	3.5 & 3.6 Referral & Discharge Summary NLP	Llama-3 8B	1000	50,000	80,000	0.120	0.72	1.15	4.18
10	4.3 Scheme Entitlement & Coverage Gap NLP Matching	BioBERT / Matcher	800	40,000	60,000	0.100	0.38	0.58	2.11
TOTAL	TOTAL 48-MONTH LLM TOKEN CONSUMPTION OUTLAY						10.43	17.04	61.55

Notes:

- Tokens are self-hosted on dedicated sovereign GPU nodes, amortized based on throughput.
- Call volumes scale smoothly from district pilots during Year 2 to statewide maturity in Years 3 through 5.

9.8 State Data Centre (SDC) Enterprise Hardware Sizing

Infrastructure sizing requirements for enterprise server nodes provided by the Government of Jharkhand at zero incremental cost.

Category	Component Role	vCPU Cores	RAM (GB)	Storage	Nodes	Cost to Project	Infrastructure Provider
Core Lake	Data Lake & ETL Cluster	128	512	50 TB NVMe	4	₹0.00 (Free)	Govt State Data Centre (SDC)
Core Identity	Master Patient Index & ABHA Database	128	256	20 TB NVMe	4	₹0.00 (Free)	Govt State Data Centre (SDC)
Core Interop	FHIR R4 Gateway & API Hub	128	256	10 TB NVMe	4	₹0.00 (Free)	Govt State Data Centre (SDC)
Core App/UI	Mobile Backend & Web Dashboards	128	256	10 TB NVMe	4	₹0.00 (Free)	Govt State Data Centre (SDC)
Use Cases 1-5	Tabular Scoring & Guideline Rules Engine	256	512	40 TB NVMe	8	₹0.00 (Free)	Govt State Data Centre (SDC)
Dev/UAT	Development & Staging Testbed	64	128	10 TB SSD	2	₹0.00 (Free)	Govt State Data Centre (SDC)

TOTAL	TOTAL SDC CPU SIZING	832 Cores	1,920 GB	140 TB NVMe/SSD	26 Nodes	₹0.00	—
--------------	-----------------------------	----------------------	---------------------	----------------------------	---------------------	--------------	----------

SDC Hosting Details:

- Hosted at the State Data Centre (Ranchi) leveraging existing government compute clusters, networking, and power infrastructure.

9.9 Operational & Field Change Management Outlay

Phased budgetary allocation for multi-source system integration, cybersecurity compliance, user capacity building, and district rollout support.

ID	Operational Budget Head	Nature	Yr 1 (₹ Cr)	Yr 2 (₹ Cr)	Yr 3 (₹ Cr)	Yr 4 (₹ Cr)	Yr 5 (₹ Cr)	Total (₹ Cr)
OP-01	Multi-System Data Integration & ETL Connectors	CapEx	2.00	0.00	0.00	0.00	0.00	2.00
OP-02	Backup, Disaster Recovery & High-Availability Operations	OpEx	0.17	0.09	0.09	0.09	0.09	0.53
OP-03	Telecom, SMS, IVR & WhatsApp Beneficiary Outreach Gateway	OpEx	0.12	0.07	0.07	0.07	0.07	0.40
OP-04	Cybersecurity, VAPT, DPDP 2023 Compliance & Responsible AI Audits	OpEx	0.50	0.20	0.20	0.20	0.20	1.30
OP-05	Statewide Capacity Building, Training & Frontline Handholding	OpEx	0.60	0.20	0.20	0.20	0.20	1.40
OP-06	Project Management Unit (PMU), Quality Assurance & M&E	OpEx	0.95	0.40	0.40	0.40	0.40	2.55
OP-07	District Rollout Support & Field Change Management	OpEx	1.13	0.35	0.35	0.35	0.35	2.54
TOTAL	TOTAL OPERATIONAL HEADS		5.47	1.31	1.31	1.31	1.31	10.72

Operational Expenditure Summary:

- OP-01 provides CapEx funding for high-throughput connectors to legacy health registries (e-Sushrut, Nikshay, HMIS, Kuber IFMS, PM-JAY).
- OP-04 covers annual mandatory third-party cybersecurity audits, CERT-In empanelled VAPT, DPDP 2023 compliance, and Responsible AI evaluations.
- OP-05, OP-06, and OP-07 support statewide frontline training, Project Management Unit oversight, and field change management across all 24 districts.

9.10 NICSI Empanelled Manpower Rate Card Reference

Standard consultancy tariffs and experience criteria established under official government procurement empanelment.

Rate Code	Experience Band	Base Rate / Mo (₹)	Margin %	Ex-GST Cost(₹)	Cost Incl-GST (₹)
R1	Consultant: 15 years and above	₹425,427.00	9.00%	₹463,715.43	₹547,184.21
R2	Consultant: 10 years to <15 years	₹364,652.00	9.00%	₹397,470.68	₹469,015.40
R3	Consultant: 6 years to <10 years	₹334,264.00	9.00%	₹364,347.76	₹429,930.36
R4	Consultant: 3 years to <6 years	₹285,644.00	9.00%	₹311,351.96	₹367,395.31
R5	Consultant: <3 years	₹255,256.00	9.00%	₹278,229.04	₹328,310.27
R6	Subject Matter Expert: 10 years to <15 years	₹486,203.00	9.00%	₹529,961.27	₹625,354.30

9.11 IndiaAI Mission GPU Subsidized Tariff Reference

Official hourly tariffs published under the IndiaAI compute portal for subsidized government cloud GPU deployments.

GPU Model	VRAM	Portal Source	Unit	Standard Rate (₹)	Subsidized Rate (₹)
NVIDIA H200 80GB (1x)	80 GB	IndiaAI Mission Compute Portal	Hour	₹450.00	₹375.00 / hr
NVIDIA H100 80GB (1x)	80 GB	IndiaAI Mission Compute Portal	Hour	₹400.00	₹320.00 / hr
NVIDIA A100 80GB (1x)	80 GB	IndiaAI Mission Compute Portal	Hour	₹320.00	₹250.00 / hr

10. Risk Assessment & Mitigation

10.1 Comprehensive Project Risk Matrix

The implementation of Artificial Intelligence within a state-wide public health ecosystem involves complex technical, clinical, operational, regulatory, and financial dependencies. To ensure proactive risk management, the Department of Health, Medical Education & Family Welfare (DoHMEFW) has formulated a comprehensive **Risk Assessment & Mitigation Register**:

Master Project Risk Register

Risk ID	Risk Category	Risk Event / Vulnerability Description	Likelihood (L)	Impact (I)	Proactive Mitigation Strategy & Institutional Safeguards
R1	Clinical Safety	AI Clinical Decision Support (CDSS) produces false-negative risk scores (e.g. missing an atypical High-Risk Pregnancy or Silicosis case), leading to adverse clinical outcomes.	Medium	High	1. Strict Human-in-the-Loop (HITL) architecture: All AI outputs are strictly advisory; no medical diagnosis or prescription is automated. 2. Multi-disciplinary Clinical AI Safety Board at RIMS conducts monthly diagnostic audits. 3. Continuous clinical override logging and algorithm threshold tuning.
R2	Data Privacy & Legal	Unauthorized access, data leakage, or non-compliance with the Digital Personal Data Protection (DPDP) Act 2023 , attracting legal penalties and eroding citizen trust.	Low	Critical	1. Enforced end-to-end cryptographic encryption (AES-256 at rest, TLS 1.3 in transit). 2. Dynamic de-identification of PII before AI model training. 3. Mandatory bi-annual VAPT audits by CERT-In empanelled auditors. 4. Fine-grained Role-Based Access Control (RBAC) and immutable audit logging.

Risk ID	Risk Category	Risk Event / Vulnerability Description	Likelihood (L)	Impact (I)	Proactive Mitigation Strategy & Institutional Safeguards
R3	Connectivity & Terrain	Cellular network blind spots in deep forest, hilly, or Naxal-affected tribal blocks prevent real-time data sync for Sahiyas and ANMs.	High	Medium	Offline-First Mobile Architecture: Sahiya and ANM apps cache patient records locally in encrypted SQLite databases and auto-sync when entering network zones. - Automated fallback to standard 2G voice telephony and SMS nudges where 4G data is unavailable.
R4	Adoption & Clinical Trust	Medical Officers and specialists resist adopting CDSS recommendations due to suspicion of "black-box" algorithms or fear of perceived clinical replacement.	Medium	Medium	Explainable AI (XAI): Models provide SHAP/LIME physiological feature explanations for every risk score. - Accredited Continuing Medical Education (CME) workshops for all government doctors. - Reinforcing that legal clinical authority and decision-making remain 100% with the treating physician.
R5	Vendor Lock-in	Dependence on proprietary AI frameworks or opaque vendor codebases prevents state ownership and escalates long-term maintenance costs.	Low	High	- Strict Open-Source-First Policy: Platform built entirely on open-source libraries (Python, PyTorch, PostgreSQL, Delta Lake, FastAPI). - Complete intellectual property (IP), source code, database schemas, and trained model weights vest exclusively with the Government of Jharkhand.
R6	Frontline Workload Fatigue	Sahiyas experience operational fatigue or confusion if AI alerts generate excessive notifications or redundant data-entry burdens.	Medium	Medium	- Intelligently throttled notification logic (maximum 3 to 5 priority task prompts daily per Sahiya). - One-click audio-guided checklists in native tribal languages to simplify visits. - Sahiya monthly incentive linkage for verified follow-up closures.
R7	Algorithmic & Tribal Bias	Predictive algorithms exhibit lower accuracy or demographic bias when applied to indigenous Scheduled Tribes and PVTGs due to unrepresentative training data.	Medium	Medium	- Rigorous demographic subgroup fairness testing across gender, ST/SC, and geography. - Mandatory inclusion of Jharkhand-specific clinical datasets from RIMS and tribal district hospitals. - Minimum parity threshold: < 5% predictive variance across demographic groups.

Risk ID	Risk Category	Risk Event / Vulnerability Description	Likelihood (L)	Impact (I)	Proactive Mitigation Strategy & Institutional Safeguards
R8	Source System Integration	Upstream state/central platforms (*HMIS, Nikshay, e-Shusrut, MASSY TMS*) alter API schemas or experience delayed data feeds, stalling ETL pipelines.	Medium	Medium	1. Standardized HL7 FHIR R4 normalization layer decoupling ingestion from internal storage. 2. Formal inter-departmental data-sharing MoUs with designated Nodal Data Officers. 3. Automated pipeline telemetry alerting SPMU engineers within 15 minutes of feed interruptions.
R9	Infrastructure & SDC Load	Unexpected spikes in concurrent user logins or data lake queries cause latency or downtime at the State Data Centre (SDC).	Low	Medium	1. Auto-scaling containerized microservices orchestrated via Kubernetes at SDC. 2. Dedicated Redis caching layers for high-frequency dashboard queries. 3. Offloading heavy model training to IndiaAI Mission cloud GPU clusters.
R10	Financial & Sustainability	Post-implementation operational funding lapses after the initial 60-month contract period, threatening platform shutdown.	Low	High	1. Multi-scheme convergence institutionalizing recurring O&M within the routine NHM PIP Digital Health budget line. 2. Substantial direct fiscal savings (₹160–200 Cr/yr) make the platform self-funding from Year 2 onwards.

11. Sustainability

11.1 Socio-Economic Impact

The proposed investment is self-sustaining and delivers immense economic and social value:

1. Reduction in preventable re-admissions:

- AI based analytics and alerts engine through timely post-discharge monitoring and proactive outpatient management will reduce the re-admission rates and will reduce annual hospital admissions under MASSY and PM-JAY are clinically preventable .

2. Disease Surveillance & Mitigation:

- Leverages continuous post-discharge data analytics and clinical anomaly detection to establish hyper-local epidemic intelligence, intercepting seasonal disease clusters at the primary healthcare level to mitigate surge burdens on district hospital infrastructure.

3. Reduction in Out-of-Pocket Expenditure (OOPE):

- Prevents catastrophic health expenditure for over 3.5 Lakh vulnerable households annually by intercepting disease complications before they necessitate private tertiary hospitalization.

4. Tribal & Social Inclusion:

- Directly bridges healthcare access barriers for 86 Lakh tribal citizens and 8 PVTG communities across remote, forested geographies through native-language voice AI and grassroots Sahiya enablement.

11.2 Post-Pilot Scale-Up & Institutionalization Strategy

A critical failure mode of public sector digital health pilots is the inability to transition from donor-funded or vendor-driven experiments into permanent, institutionalized government workflows. The **Jharkhand Health Intelligence Platform (J-HIP)** is architected from its inception to become a permanent, sovereign digital health capability of the **Department of Health, Medical Education & Family Welfare (DoHMEFW)**, Government of Jharkhand.

Institutional Mainstreaming Pillars:

1. Departmental Ownership of Health Assets:

- The State Health Data Lake, Master Patient Index, and algorithmic models are officially gazetted as sovereign public health assets owned exclusively by the Government of Jharkhand.

2. Integration into Routine Administrative Review:

- Disease Severity Index (DSI) heat-maps and post-discharge follow-up metrics are embedded as mandatory agenda items in monthly District Health Society (DHS) review meetings chaired by Deputy Commissioners.

3. Institutionalizing Frontline Tasking:

- Sahiya AI task lists are formally integrated into the state ASHA incentive claim and performance verification structure under the National Health Mission.

11.3 Technology Sustainability, AMC & Local Maintenance Capacity

To ensure technological resilience and avoid software obsolescence, J-HIP enforces strict sustainability protocols:

1. Open-Source-First Architecture:

- The platform utilizes community-supported, vendor-neutral open-source technologies (*Python, PostgreSQL, Apache Spark, Delta Lake, FastAPI, Kubernetes, Scikit-learn, PyTorch*). This guarantees that the State Government is never subjected to proprietary commercial license renewals, restrictive per-user fee models, or vendor lock-in.

2. Comprehensive Annual Maintenance Contract (AMC) Framework:

- Post completion of the 48-month O&M period included in this DPR, the Department may procure a competitive Annual Maintenance Contract (AMC) for specialized software and AI engineering, budgeted at an industry-standard costs.

3. Local Technical Capacity Building:

- Throughout Months 48–60, the System Integrator is contractually obligated to conduct structured technical shadowing and knowledge transfer for **state-designated software engineers, database administrators, and data analysts**.

11.4 Vendor Exit & Comprehensive Handover Strategy

The contract with the System Integrator mandates a **12-month formal transition and exit runway (Months 48–60)** to ensure seamless knowledge transfer and zero disruption to public health delivery:

Mandatory Handover Deliverables:

- **Full Source Code & Trained Model Binaries:** Complete, uncompiled source code, build scripts, Dockerfiles, Kubernetes manifests, and trained neural network weight files.
- **Comprehensive Technical Runbooks:** Standard Operating Procedures (SOPs) for data pipeline recovery, database backup/restore, model retraining execution, and security patch deployment.
- **Zero Encumbrance Certification:** Formal legal certification confirming that all intellectual property, data assets, and algorithms are free of third-party commercial claims and vest solely with the Government of Jharkhand.

12. Annexures

12.1 Annexure A: Stakeholder Consultation Proceedings & Field Notes

Doctors, front line workers and Administrators

Sl.No	Name and Details	Applications Currently Used	Key Pain Points Articulated
1	Dr. Jitendra Kumar MOIC CHC Gomia Gomia, Bokaro	● NCD Portal, Nikshay Portal, RCH Portal - for clinical/programme reporting ● e-Aushadhi Portal - medicine stock and near-expiry alerts ● Separate patient register - maintained manually for follow-up tracking	● "Typical OPD 70 to 80 Patient see in a day and separate register maintain." ● "Inform over phone call about the patient discharge." ● "Mainly paper they bring and reports." ● "Own Knowledge and experience." ● "Separate time for MIS reporting and separate time for patient."
2	Dr. Anil Kumar MOIC CHC Chas Chas, Bokaro	● E-Sushrut, IHIP - clinical/programme reporting portals ● OPD Register, Vaccination Register, Medicine Register - maintained in parallel as paper records ● DVDMS - used for medicine-stock and prescribing checks	● "E-Sushrut, IHIP, OPD register, Vaccination register, Medicine register etc. In the same day or it takes 1 or 2 days." ● "Feedbacks comes through ASHA workers. I take feedback through Asha workers and other front-line workers in special case." ● "By Patient reply, Prescription note and now a day by Abha history in some cases."
3	Jenny Minz & Idris Miya BPM & BDM CHC Karra Karra, Khunti	● U-WIN, HMIS, IHIP, ANMOL	● "They have to do upload same data in multiple portals, as the requirement and milestone of each and every portal is different and same data is used for multiple KPI's." ● "No system does not make ready-made reports, they have to manually make the reports as per the requirement." ● "They face issues due to regular APP updates, as they have to step by step explain it to the groundworkers, by the time they get a grasp on the app the version of the app changes and they have to again explain them everything again which causes lot of confusion." ● "They do get incomplete data sometimes which they resolve by Contacting the concerned person... they call up individually to all the respective CHO, ANM, GNM, Sahiya, wherever there is any pendency."

4	Juber Alam BDM CHC Namkum Namkum, Ranchi	● HMIS, RCH 2.0, U-WIN, PMSMA, IDSP, IHIP, Nikshay, IMCI	● “डेटा से संबंधित कार्य पूरे महीने चलता रहता है। उदाहरण के लिए, महीने की शुरुआत में HMIS से संबंधित कार्य किया जाता है। इसके बाद MDR-CDR की रिपोर्ट प्राप्त होती है और IDSP आदि से संबंधित कार्य भी उसी महीने के दौरान चलता रहता है।” ● “यदि किसी अवधि में कुल 5 जन्म हुए हैं और दूसरे पोर्टल में इन पांचों लाभार्थियों की अलग-अलग एंट्री करनी है, तो आधार नंबर या मोबाइल नंबर का मिलान नहीं होने के कारण कभी-कभी 5 में से केवल 2 या 1 लाभार्थी की ही एंट्री हो पाती है... डेटा का सत्यापन करने के लिए प्रत्येक महीने कम से कम 6-7 फील्ड विजिट की जाती हैं।”
5	Ajit Kumar BDM CHC Burmu Burmu, Ranchi	● HMIS, PMSMA, IPH, IHIP, RCH 2.0, U-WIN	● “एक ही डेटा को कई अलग-अलग पोर्टलों पर दर्ज करने के कारण डेटा एंट्री का कार्य बढ़ जाता है और फील्ड स्तर के कर्मचारियों पर अतिरिक्त कार्यभार पड़ता है।” ● “डेटा एंट्री करने के कारण प्रमुख आंकड़े हमें याद रहते हैं... जिन पोर्टलों पर हम रिपोर्ट दर्ज करते हैं, वहां से रिपोर्ट को डाउनलोड भी कर लेते हैं। डाउनलोड की गई रिपोर्ट का उपयोग डेटा की समीक्षा, विश्लेषण और आवश्यकतानुसार सत्यापन के लिए किया जाता है।” ● “सहिया से डेटा प्राप्त करने में कुछ कठिनाइयाँ आती हैं, क्योंकि कई बार उनके द्वारा डेटा अनौपचारिक या अनुमानित रूप से उपलब्ध करा दिया जाता है। इसलिए डेटा की शुद्धता सुनिश्चित करने के लिए ANM के साथ मिलकर उसका सत्यापन किया जाता है।”
6	Ashok Murmu BDM - In Charge CHC Bermo Bermo, Bokaro	● HMIS, RCH, U-WIN, E-Sushrut, MPCDSR, PMSMA, plus separate Google Sheet formats issued by district and state	● “Field staff ka data hard copy aur whatsapp ke dawara prapth hoti hai. Aisa koi system nhi hai sari entry mere level se hoti hai.” ● “AGAR AISA SYETM BANE TO KAAM BAHUT FAST HOGA.”

7	Yogendra Prasad BDM - In Charge CHC Gomia Gomia, Bokaro	● HMIS, RCH, ANMOL, U-WIN, Nikshay, NCD, e-Sanjeevani, PMSMA, e-PMSMA, IRAD (Road Accident), MCTS/PO - around a dozen separate portals	● “SARE PORTAL KA EXCESS EK JAGAH PE HO, TO USKO USS KARNE SAMAY ALAG ALAG PORTAL NAHI KHOLNA PADEGA ,OR TIME CONSUMING BHI KAM HO JAYEGA.” ● “ADIKANSH SAMAY DATA REGISTER SE DEKH KE ANM REPORT TAYAR KARTI HAI, AGAR KABHI KABHI MISSMATCH HOTA HAI TO JAB FIELD ME AAM CENTER JATA HU TO MATCH KARTA HU.”
8	Pankaj Kumar BDM CHC Chandankyari Chandankyari, Bokaro	● HMIS, U-WIN, RCH Portal, plus various Google Sheets/forms shared by State and District Headquarters.	● “Despite the availability of several modern data science tools today, reports still often need to be prepared manually in Excel. This process is time-consuming and requires considerable effort.” ● “In the HMIS reporting system, previous month's entries should remain visible while entering current month data. This would allow comparison with earlier records and help in self-assessment and validation of the data.”
9	Rambha Kumari BDM CHC Jaridih Jaridih, Bokaro	● HMIS, U-WIN, RCH & NHO, IDSP, plus Google Sheet reports from State and District Headquarters	● “रिपोर्टिंग करते समय अचानक डाटा बिना सेव के ही पोर्टल लॉग आउट हो जाता है जिससे सारा मेहनत बेकार हो जाता है और वही काम दोबारा से करना पड़ता है।” ● “एचएमआईएस की रिपोर्ट प्रणाली में पूर्व की तरह पिछले माह की प्रविष्टि भी दिखाई देनी चाहिए ताकि पिछले माह के लिए संधारित डाटा के साथ स्व मूल्यांकन किया जा सके।”
10	Chhatradhari Mahto BDM CHC Nawadih Nawadih, Bokaro	● HMIS, U-WIN, RCH Portal, plus Google Sheets/online formats shared by State and District Headquarters	● “In most portals, if an error occurs during online data entry, the system immediately displays a validation error... However, in some reporting systems, validation is not fully effective. For example, in HMIS infrastructure reporting, certain validation errors are not detected automatically.” ● “Reports often have to be prepared manually in Excel... The HMIS reporting system should display previous month's entries while entering current month data.”

11	Nilam Sarita Tirkey ANM HSC Lalkhatanga Namkum, Ranchi	● U-WIN Portal - vaccination reporting ● Janani App - pregnant-women registration and tracking ● HMIS - consolidated programme reporting	● “वर्तमान में हमारे मोबाइल फोन में ऐसा कोई ऐप या फीचर उपलब्ध नहीं है, जो यह स्वतः बताए कि आज किस लाभार्थी के घर जाना है या अगले दिन किस घर पर होम विजिट करनी है। इसलिए होम विजिट की योजना और लाभार्थियों की पहचान का कार्य मैनुअली करना पड़ता है।” ● “गर्भवती महिलाओं से संबंधित डेटा को जननी ऐप में दर्ज करते समय भी कई बार UID से संबंधित समस्याएँ आती हैं। उदाहरण के लिए, मोबाइल नंबर का मिलान नहीं होना या UID से संबंधित त्रुटि प्रदर्शित होना। इन समस्याओं के कारण कई बार गर्भवती महिलाओं का समय पर एनरोलमेंट नहीं हो पाता है।” ● “सबसे पहले उसकी एंट्री U-WIN पोर्टल में की जाती है। इसके बाद संबंधित जानकारी जननी ऐप में दर्ज की जाती है। अंत में, संबंधित डेटा HMIS में रिपोर्ट किया जाता है।” ● “हम मुंडा भाषा नहीं जानते हैं, लेकिन यदि कोई व्यक्ति केवल मुंडा भाषा समझता है और दूसरी भाषा नहीं जानता, तो उसे सहिया के माध्यम से जानकारी समझाने का प्रयास किया जाता है।”
----	---	--	---

12	Chameli Devi Mehidan Bibi, Poonam Devi, Usha Devi, Baby Taranum Sahiyas CHC Chas Chas,Bokaro	● Sahiya Mobile App and SahiyaWeb (accessed via Chrome browser) - beneficiary registration, pregnancy tracking, vaccination, and monthly reporting	● “App mein beneficiary registration, pregnancy tracking, vaccination aur reports bharte hain... Kai baar app slow chalti hai ya data sync hone mein der lagti hai. App crash bhi bahut hota hai.” ● “Ek baar operation se delivery hui thi aur mujhe 5–6 din baad pata chala. Agar pehle pata chal jaata toh main jaldi jaakar care, nutrition aur ke baare mein bata sakti thi.” ● “App ya SMS jo aate hain, unhe sab beneficiaries khud nahi samajh paate, khaaskar buzurg aur kam padh-likhe log... Agar app local bhasha mein voice message bhej sake, toh bahut fayda hoga.” ● “Beneficiary ki jaankari alag jagah jorna nahi hai aur follow up ke liye kabhi der se milti hai. ANM aur sahiyya ke records aapas mein linked ho, jisse ki koi discharge ho ya vaccination due ho, mobile phone par turant notification mil jaaye.” ● “Ek beneficiary ki jaankari kai jagah bharni padti hai, jaise register, mobile app aur monthly reporting format mein... Din bhar ki visits ke baad data entry mein lagbhag 1-2 ghante lag sakte hain.” ● “Aksar jab main ghar jaati hoon, toh puri medical history phone mein uplabdh nahi hoti. Kaafi jaankari mujhe family se poochkar ya discharge paper dekhkar leni padti hai.” ● “Agar app khud priority cases dikha de aur bata de ki kitne din ke andar visit karna hai, toh bahut upyogi hoga. Isse koi bhi high-risk case miss nahi hoga.”
----	---	--	---

13	Nivedita Kumari CHO AAM Thakurgao Burmu, Ranchi	● IDSP App - disease surveillance reporting ● NCD App, T3 App (Hb/Anaemia), Sickle Cell App, TB App - programme-specific reporting, entered separately for the same patient	● “जैसे कि अगर NCD मैंने किया, किसी patient का BP, Sugar लिया, तो उसको मैं NCD App में entry करूंगी। फिर अगर मैंने उसका HB लिया तो मुझे फिर उसे T3 App पे करना है... मतलब same patient पे अगर मैं सब कर रही हूँ तो मुझे almost around five to six apps में उनको entry करना है।” ● “हम लोग का मंथली मीटिंग होता है... तो जो हमारा टाइम है, उसमें हम लोग मंथली OPD में ही रहते हैं। कोई स्पेशल केस है तो वन-टू-टू विजिट पर मंथ होता है।” ● “नहीं-नहीं, इसका कोई मेरे पास वो app नहीं है कि मैं देख पाऊँ कि इनका वहाँ क्या treatment हुआ था। मेरे को उनसे, family से पूछना पड़ता है।” ● “Vaccination का हम लोग ये सब manual work है। ये due list हम लोग बनाते हैं every site के बाद।” ● “ऐप थोड़े कम हो जाएंगे तो अच्छा होगा हमारे लिए। अगर ये सारे एक जगह club हो जाएं, कि हमें हर अलग चीज के लिए अलग ऐप ना खोल के एक ही जगह पे अगर सब हम लोग को मिले, तो वो अच्छा होगा।”
----	--	--	---

Technology Vendors

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
1	Vendor	CDAC	PAN India_ Maintained by JAP IT	Care India	Baba Computer- Not Live	Dream Work's Infotech	Dhanush Healthcare Systems
2	Contact Person Name	Kriti Narayan	Nirmal Kumar Pahari	Aayush	Ritesh	Devanandan Oraon	Md. Irshad
3	URL	https://jh.e-sushrutnx.in/	https://pcpndt.jharkhand.gov.in/	hmis.jharkhand.gov.in	https://www.104healthlinejharkhand.in/	https://mtc.jharkhand.gov.in	https://sahiya.jharkhand.gov.in/#/ https://play.google.com/store/apps/details?id=com.sahiyaapp
4	Name of the Program	MMDHY	PC&PNDT	10bedICU		Malnutrition Treatment Centre (MTC) Programme – NHM Jharkhand	Sahiya JAP-IT Project
5	Name of Vendor (Company Name)	CDAC	Jharkhand Agency for Promotion of Information Technology (JAP-IT)	eGovernments Foundation	Baba Computers	DreamWorks Infotech Pvt. Ltd. / Current MIS Vendor	Dhanush Healthcare System Pvt. Ltd.

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
12	Description of the Application	e-Sushrut HMIS is a comprehensive Hospital Management Information System (HMIS) adopted by the Department of Health, Medical Education & Family Welfare, Government of Jharkhand. Developed by C-DAC (Centre for Development of Advanced Computing), this initiative is a major step towards realizing the vision of the Ayushman Bharat Digital Mission (ABDM). It incorporates an integrated computerized clinical information system for improved hospital administration and patient health care, providing an accurate, electronically stored medical record of the patient.	PCPNDT (Pre-Conception and Pre-Natal Diagnostic Techniques) Act regulates and monitors diagnostic procedures to prevent sex selection and female foeticide and ensures the ethical use of prenatal diagnostic techniques.	Care HMIS is an open source, FHIR compliant ABDM enabled hospital management information system with complete end to end solution hospitals being piloted in the Ramgarh district, Jharkhand.	ACVMS (Attendance Cross-Verification Management System) - HRIS Module is a web-based dashboard used to record, verify and monitor employee attendance under the 104 Health Line Jharkhand programme. Authorised users view and filter data through panels such as the Jharkhand District Map on the Dashboard (district-wise view) and View Employee Details on the View Attendance page.	Web-based MIS for monitoring MTC admissions, treatment, clinical/nutrition outcomes, discharge, referral, follow-up, bed occupancy, recovery and programme performance. The upgraded MIS is also being integrated with the SAAMAR App to enable digital referral and treatment/discharge data exchange.	Sahiya (ASHA) Application (Integrated Sahiya App Software) and mobile app is a user-friendly and versatile digital solution designed to streamline the work of Sahiya and strengthen primary healthcare services. This Application and mobile app provides a comprehensive platform accessible on both mobile and desktop devices, allowing users to access the application seamlessly and perform their tasks efficiently. It operates in both online, offline and wifi modes, ensuring uninterrupted functionality even in areas with limited connectivity.

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
13	What are the major problems faced by the user that can be resolved the AI Solution		Since the portal is currently under development, AI implementation will be planned after its completion.	Currently deployed AI Scribe(a predictive voice to text AI model) in all public health facilities in the Ramgarh district. It allows the user to add the patient health data through voice based informations. Along with this, we have generative models that generates treatment summaries and discharge summaries based on previously captured health records of the patients.	At present, users filter and control dashboard reports by manually clicking through multiple drop-downs, date pickers and selection boxes, which is slow and repetitive. The proposed AI voice solution lets authorised users apply these filters through simple spoken English commands (optimised for Indian English), reducing manual navigation time and effort.	Poor Referral from field level, despite of having contact details and information of SAM Children. Difficulty identifying and tracking of children at risk of poor treatment outcome, default/dropout or missed follow-up. Delayed identification of MTCs/districts with poor performance or low bed occupancy. Data-quality issues such as missing/invalid DOB, weight, MUAC, discharge status, referral source and follow-up information. Large volumes of data make it difficult for programme managers to identify trends and priority facilities quickly. Need for automated alerts and actionable insights rather than only descriptive dashboards. Need to	Sahiya is not so much educated that can fill the details of the beneficiary, we can give the voice based typing, just Sahiya speak all the details, system will fill all the details automatically. If Sahiya want to see any report, Sahiya will just ask to the chat bot, she will get the report. We can give option to the Sahiya to raise any grievance via voice and after solution Sahiya will get the notification with the proper solution. In any forms like home visit, pregnancy, child birth, Sahiya select the symptoms and vital, based on that AI will say that this women or child is in high risk, AI will suggest to the Sahiya any home remedy, medicine and in critical case system will suggest to go to nearest hospital with details, with map location direction. Sahiya has to manually remember and

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
						strengthen SAAMAR-MTC referral tracking and closed-loop referral monitoring.	track due dates for ANC visits, immunization, HBNC/HBYC follow-ups and MPR/incentive submission for multiple beneficiaries; AI-based smart reminders/notifications can automatically alert her on due and pending activities. Sahiya often calls the helpdesk repeatedly for simple, repetitive technical queries (login issues, form-filling doubts); an AI chatbot can resolve these instantly, reducing helpdesk load and resolution time.
14	Are you Planning to introduce AI Solution in your application under current work order (Yes/No)		Since the portal is currently under development, AI implementation will be planned after its completion.	The AI solution is already active.	Yes	Yes – AI integration may be introduced as a decision-support layer within the existing MTC Online MIS, subject to approval, technical feasibility, data-quality assessment, privacy/security safeguards and availability under the current work order.	Yes

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
15	What Functionalities are you planning to use proposed AI for, Please Describe		Since the portal is currently under development, AI implementation will be planned after its completion.	Currently deployed AI Scribe(a predictive voice to text AI model) in all public health facilities in the Ramgarh district. It allows the user to add the patient health data through voice based informations. Along with this, we have generative models that generates treatment summaries and discharge summaries based on previously captured health records of the patients.	A voice-activated control gateway (MVP) integrated into two dashboard panels -the Jharkhand District Map on Dashboard (JDMoD) and View Employee Details on View Attendance (VEDoVA). A user speaks a command; the system transcribes the speech, uses a Generative AI / LLM to extract the intended filter parameters as structured JSON, and applies them to the existing dashboard filters. Ambiguous or out-of-scope commands trigger a fallback pop-up showing default filters and support contact numbers. (This phase excludes text-to-speech, regional languages / Hinglish, and any direct database or SQL access.)	Proposed priority functionalities: AI-based performance/risk dashboard for MTCs, districts and facilities. Predictive risk flagging for likely default, poor outcome or missed post-discharge follow-up. Automated identification of abnormal/inconsistent/missing data and duplicate records. AI-generated programme summaries highlighting major gaps, improvements and priority facilities. Automated alerts for overdue follow-up, prolonged stay, low bed occupancy, low recovery, high default or unusual trends. SAAMAR-MTC referral tracking with alerts for pending referrals and incomplete referral closure. Natural-language MIS assistant for programme managers to ask questions such as "Which MTCs have low recovery this month?" or "Which facilities	Voice-based data entry (Speech-to-Text) across major forms such as Family Registration, EC, ANC, Pregnant Woman, Delivery, Child Birth and Home Visit forms. AI Chatbot for retrieving reports and dashboards on voice/text query, and for voice-based grievance registration with automated status notification. AI-based risk assessment on symptoms/vital signs captured in ANC, Delivery, Child Birth and Home Visit forms, to flag high-risk mother/child cases automatically. AI-generated home-remedy and care suggestions for low-risk cases, and nearest-hospital referral with map-based direction for high-risk/critical cases. Smart reminder/notification engine for due follow-

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
						have high pending follow-up?" AI-assisted generation of district/state review briefs and action points. Trend/anomaly detection for admissions, referrals, outcomes, occupancy and follow-up. AI must support programme decision-making and not replace clinical judgement. AI Chat-bot can be introduced for beneficiaries and service providers	ups, immunization schedules and incentive submission deadlines.
16	What are the timelines for pilot implementation of proposed AI Solutions		Since the portal is currently under development, AI implementation will be planned after its completion.	Already deployed	18 weeks (90 working Days)	8-12 weeks after approval, data-access/API readiness and finalisation of use cases.	Month 1: Requirement finalization and selection of AI/LLM API partner; data privacy and consent framework approval. Month 2: Development and integration of voice-based data entry and chatbot module for selected forms. Month 3: Development of AI-based risk assessment logic for priority forms (ANC, Delivery, Child Birth). Month 4: Pilot rollout in 1-2

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
							selected blocks/districts with a limited group of Sahiyas. Month 5: Field feedback collection, accuracy review and refinement of AI suggestions. Month 6: Pilot evaluation report and readiness assessment for state-wide scaling.
17	What support required for Scaling in Pilot		Since the portal is currently under development, AI implementation will be planned after its completion.	Since the state has the ownership of the platform, the most important support would be bearing the costs implicated for the use of said AI models.	A designated HRIS Nodal Officer to coordinate implementation ; agreed SOPs, rules and regulations for HRIS; and a committee to define and approve the scope of work.	Representative MTC/district pilot sites and programme focal persons. Clean historical MTC data for model development/validation. API/data access for MTC MIS and SAAMAR integration. Defined indicator dictionary, business rules and outcome definitions. IT/vendor support for integration, testing and troubleshooting. User acceptance testing with programme and MTC users.	Selection and approval of 1-2 pilot districts/blocks with reasonably stable internet connectivity. Provision/upgradation of smartphones with adequate specification for Sahiyas participating in the pilot, wherever required. Approval and budget for cloud-based AI/LLM API usage during the pilot phase. Nomination of a nodal officer from the department for pilot monitoring, field

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
						Training/orientation for identified State, District and MTC users.	coordination and feedback collection. Support in collecting sample multilingual (Hindi/local dialect) voice data for improving speech-to-text accuracy.
18	What support required for Scaling across State		Since the portal is currently under development, AI implementation will be planned after its completion.	Since the state has the ownership of the platform, the most important support would be bearing the costs implicated for the use of said AI models.	Dedicated day-to-day manpower - State/district-level nodal officers and IT support staff to handle rollout coordination, user onboarding, and issue escalation across districts. Training and change-management support to onboard district-level staff and end users to the new voice-based feature.	State-level hosting/infrastructure and required compute capacity. Stable APIs and integration with SAAMAR and other approved programme systems. Statewide data-quality and standardisation mechanism. Role-based access, audit trail, encryption and security controls. State-level AI dashboard and alerting mechanism. Training and change-management support. Periodic model validation, monitoring and	State-level budget approval for AI/LLM API and compute costs at full scale (all 24 districts). Improvement of network/internet infrastructure in remote blocks to support voice-based and AI features reliably. Device upgrade support for Sahiyas currently using low-configuration smartphones or without smartphones. Data privacy and security clearance from the department for AI processing of health-related data.

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
						recalibration. Dedicated technical support for AI/MIS maintenance and integration	State-wide training and change-management support to help Sahiyas adopt AI-based features confidently. Integration support from the department for cross-system data (e.g. NCD, U-WIN APIs) to strengthen AI-based risk assessment.
19	Which AI/ML Technology are you planning to use for the proposed AI Solutions (Please Tick the relevant)	Generative AI & LLM		Generative AI & LLM	Generative AI & LLM	☒ Generative AI & LLM – MIS conversational assistant, automated summaries, review notes and programme queries.	
		Predictive AI & ML		Predictive AI & ML		☒ Predictive AI & ML – risk prediction for default/follow-up, outcome trends, performance forecasting and anomaly detection.	
		Computer Vision & Multimodal AI					

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
		AI Agents & Automation				☒ AI Agents & Automation – automated alerts, data-quality checks, referral follow-up and workflow-based notifications.	
		Decision Intelligence & Optimization				☒ Decision Intelligence & Optimization – facility prioritisation, performance review, resource/bed-utilisation insights and programme action planning.	
20	Requirement/Support from State for Implementation of AI Solutions		Since the portal is currently under development, AI implementation will be planned after its completion.	Since the state has the ownership of the platform, the most important support would be bearing the costs implicated for the use of said AI models.	Day-to-day manpower support for coordination and monitoring during the pilot. A designated Nodal Officer / SPOC from the State side to coordinate approvals, testing, and sign-offs.	Approval of AI use cases and pilot scope by NHM Jharkhand. Access to required historical and current MIS data, with appropriate privacy and security safeguards. SAAMAR–MTC API readiness, including DOB and referral/discharge workflow requirements. Defined data dictionary, indicator definitions, business rules and outcome classifications. Support from State IT/JAP-IT	Not much requirements on AI

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
						for hosting, API, security, access and infrastructure requirements. Nomination of programme and IT focal persons for AI implementation and validation. Security, privacy, audit and role-based access assessment before production deployment. Provision for pilot, model validation	
21	Improvement Required in existing Application	Chat Bot & ePrescription Automation	“The currently operational portal is experiencing performance issues, including slow response times and data duplication. To address these issues, the portal is being redeveloped from the ground up to improve performance, enhance data integrity, and resolve all reported issues to date.”		Front-end enhancements to the existing ACVMS dashboard: add a microphone button, integrate the browser Web Speech API for speech-to-text, add a JavaScript listener to receive the AI gateway's JSON filter payload, programmatically update the existing jQuery filters and refresh the panels, and add a fallback pop-up for ambiguous commands.	Improve data validation at entry level with mandatory/conditional fields and logical checks. Strengthen duplicate detection and unique child identification. Complete SAAMAR-MTC referral and discharge data exchange with closed-loop referral status. Add automated alerts for pending referral, admission, discharge and post-discharge follow-up. Improve	Voice AI – The Sahiya does not fill the forms manually. The forms get filled by voice recognition. AI Chatbot – To generate the required reports anytime without the help of technical team HRPW – High risk prediction

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
						dashboard filters by State/District/Block/MTC, period, age/sex and referral source.	
22	GPU and AI Infrastructure Considerations	For pilot implementations, cloud-based AI services are expected to be sufficient. However, state-wide deployment of advanced analytics, clinical decision support, and large-scale AI-assisted documentation may require centralized GPU-enabled infrastructure hosted at the Jharkhand State Data Centre (SDC) or a Government Community Cloud environment.	No immediate GPU requirement has been identified. Future use cases such as anomaly detection, compliance monitoring, fraud detection, and inspection prioritization may require AI processing resources.	While current deployments utilize AI services successfully, large-scale statewide adoption may increase computational demand. Jharkhand may consider establishing centralized GPU infrastructure or shared AI services to optimize long-term operational costs and improve scalability.	The proposed use cases are lightweight and can be supported through cloud-hosted AI APIs. No dedicated GPU infrastructure is expected to be required during the pilot phase.		Voice-enabled services, multilingual support, and large-scale beneficiary risk assessment may require centralized AI processing capability. A shared GPU infrastructure hosted at the State Data Centre or cloud-based AI services is recommended for statewide implementation.

Sl. No.	Name of the State Own Application	e Shusrut (C-DAC)- ABDM Cell	Garima Jharkhand (State Initiative) / PCPNDT	HMIS (Care Portal)	HRIS (Human Resource Information System)	Malnutrition Treatment Center MIS (State Initiative)/NRC - Child Health Cell	Sahiya Portal and App (State Initiative) / ASHA cell
23	AI Opportunities	AI-assisted clinical documentation. Automated discharge summary generation. Patient query and support chatbot. Predictive hospital utilization analytics. Clinical decision-support tools.	Duplicate record identification. Compliance violation prediction. Intelligent inspection prioritization. Automated compliance monitoring and alerts.		Conversational HR analytics. Voice-enabled dashboard navigation. Intelligent employee information retrieval. Predictive workforce planning and vacancy analysis.		

12.2 Annexure B: Suggested Service Level Agreement (SLA)

Governance Principles

The SLA framework is designed in alignment with MeitY AI Governance Guidelines 2025, GFR 2017, and industry best practices for Government AI/IT programmes. The following principles govern this SLA:

- Performance-linked payments: A minimum 15% of quarterly SI payments shall be at risk, contingent on SLA performance
- Balanced accountability: SLA obligations are binding on both SI and DoH&FW (including timely data access, approvals)
- Continuous improvement: SLA targets may be enhanced during contract reviews, not reduced
- Transparency: All SLA metrics published on a live dashboard accessible to PMU, PSC, and authorised stakeholders

Platform Availability & Performance SLAs

SN	Service Component	Target SLA	Minimum Acceptable	Penalty / Consequence
1	Core Platform Uptime (Production)	99.5% monthly	99.0%	< 99.5%: 1% deduction per 0.1% shortfall; < 99.0%: 3% deduction
2	Critical AI Services (CDSS, Outbreak Alert)	99.9% uptime	99.5%	< 99.9%: 2% deduction per 0.1% shortfall; < 99.5%: 5% deduction
3	Dashboard Response Time (95th Percentile)	< 3 seconds	< 5 seconds	Breach for 2 consecutive months: 2% quarterly deduction
4	API Response Time (Health data exchange)	< 500ms	< 1 second	Breach for 2 consecutive months: 1.5% deduction
5	Batch Data Processing (Nightly ETL)	Completion by 06:00 IST daily	By 08:00 IST	Failure > 3 days/month: 1% deduction
6	AI Model Inference Latency (Real-time)	< 2 seconds per prediction	< 5 seconds	Breach on critical clinical modules: escalation to TRC + 2% deduction
7	Disaster Recovery RTO	4 hours (Category A systems)	8 hours	Each breach: 5% penalty capped at 10% per quarter
8	Disaster Recovery RPO	1 hour	4 hours	Each breach: 3% penalty

Incident Management SLAs

Priority	Definition & Examples	Initial Response	Workaround	Resolution	SLA Breach Penalty
P1 Critical	Complete outage of production platform; Outbreak alert failure; Data breach; CDSS failure at any facility	15 minutes (24x7)	2 hours	4 hours	₹50,000 per hour beyond resolution time; Escalation to PSC
P2 High	Major module degradation; Significant performance drop; Batch job failure; Integration failure with ABDM	30 minutes	4 hours	8 hours	₹20,000 per hour beyond resolution time
P3 Medium	Non-critical module issue; Performance degradation; Reporting errors; AI model accuracy drop	2 hours	8 hours	48 hours	₹5,000 per incident beyond SLA
P4 Low	Minor bugs; UI issues; Training requests; Enhancement queries	8 business hours	N/A	10 business days	Tracked; repeat breach triggers TRC review

AI Model Performance SLAs

All AI/ML models deployed in production shall meet the minimum performance standards. Models failing to meet these thresholds shall be retrained/replaced within the timelines specified. The AI models have demonstrable performance measured through Sensitivity, Specificity, MAPE, False Positive/Negative rate and AUC and other indicators as applicable for the use case.

For use cases where there is predictive modeling is involved, the vendor shall declare the weights used for modeling and provide justification of selection of such weights.

12.3 Annexure C: List of Abbreviation and Acronyms

Abbreviation	Full Terminology
ABDM	Ayushman Bharat Digital Mission
ABHA	Ayushman Bharat Health Account (14-digit Unique Health ID)
AI / ML	Artificial Intelligence / Machine Learning
ANC / PNC	Antenatal Care / Postnatal Care
ANM	Auxiliary Nurse Midwife
API	Application Programming Interface
ASHA	Accredited Social Health Activist (termed *Sahiya* in Jharkhand)
ASR / TTS	Automatic Speech Recognition / Text-to-Speech
BCR	Benefit-Cost Ratio
BoQ	Bill of Quantities
CBA	Cost-Benefit Analysis
CDSS	Clinical Decision Support System
CHC	Community Health Centre
CHO	Community Health Officer
CKD / CVD	Chronic Kidney Disease / Cardiovascular Disease
COPD	Chronic Obstructive Pulmonary Disease
CWP	Coal Workers' Pneumoconiosis (Black Lung Disease)
DHS	District Health Society
DoHMEFW	Department of Health, Medical Education & Family Welfare
DPDP Act	Digital Personal Data Protection Act, 2023
DSI	Disease Severity Index
DVDMS	Drugs & Vaccine Distribution Management System (e-Aushadhi)
EHR / EMR	Electronic Health Record / Electronic Medical Record
ETL	Extract, Transform, and Load
FHIR	Fast Healthcare Interoperability Resources (HL7 Standard)
FLW	Frontline Health Worker (Sahiya, Sahiya Sathi, ANM, MPW)
HCRM	Health Compliance & Risk Management Module
HFR / HPR	Health Facility Registry / Healthcare Professionals Registry

Abbreviation	Full Terminology
HITL	Human-in-the-Loop Clinical Oversight
HMIS	Health Management Information System
HRP	High-Risk Pregnancy
HWC / SC	Ayushman Arogya Mandir / Health & Wellness Centre / Sub-Centre
ICCC	Integrated Command and Control Centre
IMR / MMR	Infant Mortality Rate / Maternal Mortality Ratio
IRR	Internal Rate of Return
J-HIP	Jharkhand Health Intelligence Platform
MASSY	Mukhyamantri Abua Swasthya Suraksha Yojana (State Cover up to ₹15 Lakh)
MEL	Monitoring, Evaluation, and Learning
MeitY	Ministry of Electronics and Information Technology, Government of India
MO / MOIC	Medical Officer / Medical Officer In-Charge
MPI	Master Patient Index
NCD	Non-Communicable Disease (Hypertension, Diabetes, Cancer, etc.)
NFHS-5	National Family Health Survey - 5 (2019–21)
NICSI	National Informatics Centre Services Inc.
NTEP	National Tuberculosis Elimination Programme (Nikshay)
O&M	Operations and Maintenance
OOPE	Out-of-Pocket Expenditure
PDCM	Post-Discharge Care Management Module
PHC	Primary Health Centre
PHM	Population Health Management
PIP	Programme Implementation Plan (National Health Mission)
PM-JAY	Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana
PMU / SPMU	Project Management Unit / State Project Management Unit
PVTG	Particularly Vulnerable Tribal Group (8 groups in Jharkhand)
RCH	Reproductive and Child Health
RHS	Rural Health Statistics Bulletin
RIMS	Rajendra Institute of Medical Sciences, Ranchi
SDC	Jharkhand State Data Centre, Ranchi

Abbreviation	Full Terminology
SDG	Sustainable Development Goals (United Nations)
SDH / DH	Sub-Divisional Hospital / District Hospital
SI	System Integrator
SNCU	Special Newborn Care Unit
SOP	Standard Operating Procedure
SRS	Sample Registration System (Office of the Registrar General of India)
ToT	Training of Trainers
VAPT	Vulnerability Assessment and Penetration Testing
VfM	Value for Money

12.4 Annexure G: Government Schemes & API References

Government Schemes

Scheme / API / Platform	Reference URL
ABDM / ABHA APIs	https://abdm.gov.in/
eSanjeevani Teleconsultation	https://esanjeevani.mohfw.gov.in/
Bhashini NLP / Translation API	https://bhashini.gov.in/
IndiaAI Mission	https://indiaai.gov.in/
PM-JAY / NHA Dashboard	https://dashboard.pmjay.gov.in/
DVDMS Drug Supply Management	https://dvdms.gov.in/
IDSP Surveillance Portal	https://idsp.mohfw.gov.in/
CPGRAMS Grievance Portal	https://pgportal.gov.in/
C-DAC e-Hospital	https://ehospital.gov.in/
India TB Report / NTEP	https://tbcindia.gov.in/
MoHFW / RHS Data	https://main.mohfw.gov.in/
Census / SRS Data	https://censusindia.gov.in/

REFERENCES

The consolidated reference list below is provided for reference convenience.

Source / Document	URL
SRS Statistical Report 2024	https://censusindia.gov.in/nada/index.php/catalog/47152
SRS MMR Bulletin 2022-24	https://censusindia.gov.in/nada/index.php/catalog/47151
NFHS 6 Fact Sheets	https://rchiips.org/nfhs/
NFHS-5 Jharkhand Fact Sheet	https://rchiips.org/nfhs/NFHS-5Reports/Jharkhand.pdf
NFHS-5 District Fact Sheets (all districts)	https://rchiips.org/nfhs/
Rural Health Statistics 2021-22 (RHS)	https://main.mohfw.gov.in/sites/default/files/RHS%202021-22.pdf
Census 2011 - Jharkhand	https://censusindia.gov.in/
India TB Report 2024	https://tbcindia.gov.in/
WHO Report - Continuity and coordination of care	https://www.who.int/publications/i/item/9789241514033
Global burden of postoperative death	https://pubmed.ncbi.nlm.nih.gov/30722955/
Global Burden of Disease 2021	https://pubmed.ncbi.nlm.nih.gov/37331373/
NHSRC Health Dossier - Jharkhand	https://nhsrccindia.org/
IndiaAI Mission (MeitY, GoI Cabinet March 2024)	https://indiaai.gov.in/
NHA / PM-JAY Dashboard	https://dashboard.pmjay.gov.in/
India State of Forest Report	https://fsi.nic.in/forest-report-2023
Rajya Sabha Unstarred Question No. 1403	https://sansad.in/rs/questions/questions-and-answers
Ayushman Bharat Digital Mission (ABDM)	https://abdm.gov.in/

eSanjeevani Teleconsultation Portal	https://esanjeevani.mohfw.gov.in/
Bhashini - Indian Language NLP API	https://bhashini.gov.in/
DVDMS Drug Supply Management	https://dvdms.gov.in/
IDSP Disease Surveillance Portal	https://idsp.mohfw.gov.in/
CPGRAMS Grievance Portal	https://pgportal.gov.in/
C-DAC e-Hospital Platform	https://ehospital.gov.in/

Regulatory References

- Digital Personal Data Protection Act (DPDP) 2023 - governs collection, processing, and storage of personal health data
- CDSCO AI/ML-Based Medical Device Guidelines - applicable to AI modules used in clinical decision support
- ABDM Data Exchange Standards - FHIR R4, HL7 for interoperability with national digital health infrastructure
- IT Act 2000 (amended) - cybersecurity obligations and data breach notification requirements
- ISO 27001 - information security management framework for the AI platform
- Central Vigilance Commission (CVC) guidelines - applicable to all public procurement processes